Introduction to Analysis II

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Abstract

Lecture note of Introduction to Analysis II.

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Chapter 1

Several Variable Differential Calculus

Lecture 1

In this chapter, we want to approximate non-linear functions by linear maps. If we consider 24 Feb.

$$f : \mathbb{R}^n \to \mathbb{R}^m \quad f(\underbrace{x_1, x_2, \ldots, x_n}_{x}) = (f_1(x), f_2(x), \ldots, f_m(x)),$$

where $f_i : \mathbb{R}^n \to \mathbb{R}$ for all i. Now given a real-valued function $F : \mathbb{R}^n \to \mathbb{R}$ which is differentiable at a point x_0. We know that near $x_0 \in \mathbb{R}^n$ we can approximate $F(x)$ in the following way:

$$F(x) \approx F(x_0) + \nabla F(x_0) \cdot (x - x_0)$$

where

$$\nabla F(x_0) = \left(\frac{\partial F(x_0)}{\partial x_1}, \frac{\partial F(x_0)}{\partial x_2}, \ldots, \frac{\partial F(x_0)}{\partial x_n} \right) \in \mathbb{R}^n \text{ with } x_0 = (x_1, x_2, \ldots, x_n)$$

and thus

$$\begin{aligned} \nabla F(x_0) \cdot (x - x_0) &= \left\langle \frac{\partial F(x_0)}{\partial x_1}, \frac{\partial F(x_0)}{\partial x_2}, \ldots, \frac{\partial F(x_0)}{\partial x_n} \right\rangle \cdot \langle x_1, x_2, \ldots, x_n \rangle \\ &= \sum_{i=1}^{n} \frac{\partial F(x_0)}{\partial x_i} x_i. \end{aligned}$$

Hence,

$$f(x) = \begin{pmatrix} f_1(x_0) \\ f_2(x_0) \\ \vdots \\ f_n(x_0) \end{pmatrix} \approx \begin{pmatrix} f_1(x_0) + \nabla f_1(x)(x - x_0) \\ f_2(x_0) + \nabla f_2(x)(x - x_0) \\ \vdots \\ f_n(x_0) + \nabla f_n(x)(x - x_0) \end{pmatrix},$$

which gives

$$f(x) - f(x_0) \approx \begin{pmatrix} \nabla f_1(x_0)(x - x_0) \\ \nabla f_2(x_0)(x - x_0) \\ \vdots \\ \nabla f_n(x_0)(x - x_0) \end{pmatrix} = \begin{pmatrix} \nabla f_1(x_0) \\ \nabla f_2(x_0) \\ \vdots \\ \nabla f_n(x_0) \end{pmatrix} \cdot \underbrace{(x - x_0)}_{\text{column vector}}.$$

1.1 Linear Transformation

Definition 1.1.1 (Row vectors). Let $n \geq 1$ be an integer. We refer to elements of $\mathbb{R}^n$ an n-dimensional row vectors. A typical row vector is $x = (x_1, x_2, \ldots, x_n)$ which we abbreviate as $(x_i)_{1 \leq i \leq m}$. The components $x_1, x_2, \ldots, x_n$ are real numbers. If x and y are two row vectors in $\mathbb{R}^n$, we can define vector sum by

$$x + y = (x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n) := (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n).$$

If $c \in \mathbb{R}$ is any real number, we define scalar multiplications by

$$cx = c(x_1, x_2, \ldots, x_n) = (cx_1, cx_2, \ldots, cx_n).$$

Remark 1.1.1.

(1) $-x := (-1) \cdot x = (-x_1, -x_2, \ldots, -x_n)$.

(2) zero vector is denoted by 0, i.e. $(0, 0, \ldots, 0)$.

Lemma 1.1.1 ($\mathbb{R}^n$ is a vector space). Let x, y, z be vectors in $\mathbb{R}^n$, and let $c, d \in \mathbb{R}$. Then the following properties hold:

(a) $x + y = y + x$.

(b) $(x + y) + z = x + (y + z)$.

(c) $x + 0 = 0 + x = x$.

(d) $x + (-x) = (-x) + x = 0$.

(e) $(c \cdot d)x = c \cdot (dx)$.

(f) $c(x + y) = cx + cy$.

(g) $(c + d)x = cx + dx$.

(h) $1x = x$.

Definition 1.1.2. Let $x = (x_1, x_2, \ldots, x_n)$ be row vector. Its transpose is the n-dimensional column vector

$$x^T = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Definition 1.1.3. The standard basis of $\mathbb{R}^n$ consists of $e_1, e_2, \ldots, e_n$, where e_j has 1 in the j-th position and 0 elsewhere:

$$e_j = (0, \ldots, 0, \underbrace{1}_{j\text{-th}}, 0, \ldots, 0).$$

Every row vector

$$x = (x_1, x_2, \ldots, x_n) = \sum_{j=1}^{n} x_j e_j.$$

Similarly,

$$x^T = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \sum_{j=1}^{n} x_j e_j^T.$$

Definition 1.1.4 (Linear transformation). A linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is any function from one Euclidean space to another that satisfies the following two properties:

(a) Additivity: For $x, y \in \mathbb{R}^n$, $T(x+y) = T(x) + T(y)$.

(b) Homogenity: For $x \in \mathbb{R}^n$ and all scalars $c \in \mathbb{R}$, $T(cx) = cT(x)$.

Remark 1.1.2. This definition is equivalent to the following:

$$T(c_1v_1 + \cdots + c_kv_k) = c_1T(v_1) + \cdots + c_kT(v_k)$$

where $v_1, \ldots, v_k \in \mathbb{R}^n$ and $c_1, c_2, \ldots, c_n \in \mathbb{R}$.

Definition 1.1.5. Let $m, n \geq 1$ be integers. An $m \times n$ ordered matrix is an ordered rectangular array of real numbers

$$A = (a_{ij})_{\substack{1\leq i\leq m\\ 1\leq j\leq n}} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

consisting of m rows and n columns, where

(a) The entry a_{ij} denote the number in the i-th row and j-th column.

(b) We denote the set of all $m \times n$ matrices by $\mathbb{R}^{m\times n}$.

(c) In particular, a row vector is a $1 \times n$ matrix, a column vector is a $n \times 1$ vector.

Definition 1.1.6 (Matrix multiplication). Given an $m \times n$ matrix $A = (a_{ij})_{\substack{1\leq i\leq m\\ 1\leq j\leq n}}$ and an $n \times p$ matrix $B = (b_{jk})_{\substack{1\leq j\leq n\\ 1\leq k\leq p}}$, we define AB to be the $m \times p$ matrix $(c_{ik})_{\substack{1\leq i\leq m\\ 1\leq k\leq p}}$ where

$$c_{ik} = \sum_{j=1}^{n} a_{ij}b_{jk}.$$

Definition 1.1.7 (Matrix-vector multiplication). Let $A = (a_{ij}) \in \mathbb{R}^{m\times n}$ and $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n$ be a column vector. We define

$$Ax = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n \end{pmatrix}$$

Remark 1.1.3. In our class, we just treat $\mathbb{R}^n, \mathbb{R}^m$ as column vector spaces, and $L_A(x) = Ax$ is a $m \times 1$ column vector.

Theorem 1.1.1. Let A be a $m \times n$ matrix, then $L_A(x) = Ax$ is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Proof. ■ DIY

Proposition 1.1.1. Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. For each $j = 1, 2, \ldots, n$, let e_j denote the j-th standard basis vector in $\mathbb{R}^n$ and write $T(e_j) = \begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{pmatrix}$. Define the matrix $A = (a_{ij})$, then $T(x) = Ax$.

Proof. Let $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$. We can write $x = \sum_{j=1}^n x_j e_j$, then we know

$$T(x) = T\left(\sum_{j=1}^n x_j e_j\right) = \sum_{j=1}^n x_j T(e_j) = \sum_{j=1}^n x_j \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix} = Ax.$$

■

Lemma 1.1.2. Let A be a $m \times n$ matrix and let B be a $n \times p$ matrix. Then $L_A \circ L_B = L_(AB)$.

Proof. It suffices to show that $(L_A \circ L_B)(x) = L_{AB}(x)$ for $x \in \mathbb{R}^p$, and the rest is easy. ■

As previously seen. $f : E \to \mathbb{R}$ where E is a subset of $\mathbb{R}$, then

$$f'(x_0) = \lim_{\substack{x \to x_0 \\ x \in E \setminus \{x_0\}}} \frac{f(x) - f(x_0)}{x - x_0}.$$

Suppose now $f : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$, we can't define

$$f'(x) = \lim_{\substack{x \to x_0 \\ x \in E \setminus \{x_0\}}} \frac{f(x) - f(x_0)}{x - x_0}$$

since the denominator and the numerator are vectors in $\mathbb{R}^n$ and $\mathbb{R}^m$.

1.2 Derivatives in Several Variable Calculus

Lemma 1.2.1. Let $E \subseteq \mathbb{R}$, let $f : E \to \mathbb{R}$ be a function and let $L \in \mathbb{R}$ and x_0 be a limit point of E. Then the following two statements are equivalent:

(a) f is differentiable at x_0 and $f'(x_0) = L$.

(b) $\lim_{\substack{x \to x_0 \\ x \in E \setminus \{x_0\}}} \frac{|f(x) - f(x_0) - L(x - x_0)|}{|x - x_0|} = 0$.

Proof. Note that

$$\frac{f(x) - f(x_0)}{x - x_0} = L + \frac{f(x) - f(x_0) - L(x - x_0)}{x - x_0} \text{ if } x \neq x_0,$$

so we have

$$\frac{f(x) - f(x_0)}{x - x_0} - L = \frac{f(x) - f(x_0) - L(x - x_0)}{x - x_0} \text{ if } x \neq x_0,$$

and thus

$$0 = \lim_{\substack{x \to x_0 \\ x \in E \setminus \{x_0\}}} \left| \frac{f(x) - f(x_0)}{x - x_0} - L \right| = \lim_{\substack{x \to x_0 \\ x \in E \setminus \{x_0\}}} \left| \frac{f(x) - f(x_0) - L(x - x_0)}{x - x_0} \right|.$$

■

Lecture 2

26 Feb.

Definition 1.2.1 (Differentiability). Let E be a subset of $\mathbb{R}^n$, let $f : E \to \mathbb{R}^m$ be a function, and let x_0 be a limit point of E. Let $L : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. We say f is differentiable at x_0 with derivative L if

$$\lim_{\substack{x \to x_0 \\ x \in E \setminus \{x_0\}}} \frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = 0,$$

or equivalently, given $\varepsilon > 0$, $\exists \delta > 0$ s.t. for all $x \in E$ satisfying $0 < \|x - x_0\| < \delta$, we have

$$\frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} < \varepsilon.$$

Remark 1.2.1. $\|f(x) - f(x_0) - L(x - x_0)\|$ is the length of a vector in $\mathbb{R}^m$ and $\|x - x_0\|$ is the length of a vector in $\mathbb{R}^n$.

Remark 1.2.2. x_0 is a limit point of $E \subseteq \mathbb{R}^n$ if for any $r > 0$, $B(x_0, r) \cap (E \setminus \{x_0\}) \neq \varnothing$. That is, for every $r > 0$, $\exists x \in E$ s.t. $0 < \|x - x_0\| < r$.

Example 1.2.1. Suppose $f : \mathbb{R}^n \to \mathbb{R}^n$ has

$$f(x) = \begin{cases} 0, & \text{if } x \in E = \{(x_1, x_2, \ldots, x_n) \mid x_n \geq 0\}; \\ x, & \text{if } x \in \mathbb{R}^n \setminus E \end{cases}$$

then f is differentiable on E and $f'(x) = 0$ and f is not differentiable on $\mathbb{R}^n$ at 0. This is because on E, f is a constant function, so

$$\frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = 0$$

for $x_0 \in E, x \in E \setminus \{x_0\}$. On $\mathbb{R}^n$ and $x_0 = 0$, then for $x \in \mathbb{R}^n \setminus \{x_0\} = \mathbb{R}^n \setminus \{0\}$, then if we approach to 0 along E, then it gives

$$\lim_{\substack{x \to x_0 \\ x \in E}} \frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = 0,$$

but if we approach 0 along $x_n < 0$, then it becomes

$$\lim_{\substack{x \to x_0 \\ x \in \mathbb{R}^n \setminus E}} \frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = \lim_{\substack{x \to x_0 \\ x \in \mathbb{R}^n \setminus E}} \frac{\|x - 0 - L(x - x_0)\|}{\|x\|} = 1$$

for any linear map L since any linear map is differentiable and thus continuous. Thus, f is not differentiable at 0 on $\mathbb{R}^n$.

Remark 1.2.3. Recall that x_0 is said to be an interior point of $E \subseteq \mathbb{R}^n$ if there exists $r > 0$ s.t.

$$B(x_0, r) = \{x \in \mathbb{R}^n : \|x - x_0\| < r\} \subseteq E.$$

If x_0 is an interior point of E, then f is differentiable at x_0 is equivalent to

$$\lim_{\substack{h \to 0 \\ x_0 + h \in E}} \frac{\|f(x_0 + h) - f(x_0) - L(h)\|}{\|h\|} = 0$$

since if $\|h\| < r$, then $x_0 + h \in B(x_0, r)$, which gives $x_0 + h \in E$, and thus $f(x_0 + h)$ is well-defined.

Remark 1.2.4. Here $\|\cdot\|$ denoted the standard Euclidean norm on $\mathbb{R}^n$ (and on $\mathbb{R}^m$):

$$\|(x_1, x_2, \ldots, x_n)\| = \left(x_1^2 + x_2^2 + \cdots + x_n^2\right)^{\frac{1}{2}}.$$

Example 1.2.2. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be given by

$$f(x, y) = \left(x^2, y^2\right),$$

let $x_0 = (1, 2)$, and let $L : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear map

$$L(x, y) = (2x, 4y).$$

We claim that f is differentiable at x_0 with derivative L.

Proof. By definition, f is differentiable at x_0 with derivative L if and only if

$$\lim_{h \to 0} \frac{\|f(x_0 + h) - f(x_0) - L(h)\|}{\|h\|} = 0,$$

where $h = (a, b)$. Thus, we know the above equation becomes

$$\frac{\|f(1 + a, 2 + b) - f(1, 2) - L(a, b)\|}{\|(a, b)\|} = \frac{\sqrt{a^4 + b^4}}{\sqrt{a^2 + b^2}},$$

and since

$$0 \leq \frac{\sqrt{a^4 + b^4}}{\sqrt{a^2 + b^2}} \leq \sqrt{a^2 + b^2},$$

so by the Squeeze Theorem, we have

$$\lim_{(a,b) \to (0,0)} \frac{\sqrt{a^4 + b^4}}{\sqrt{a^2 + b^2}} = 0,$$

and we're done. ⊛

Lemma 1.2.2 (Uniqueness of derivative)**.** Let E be a subset of $\mathbb{R}^n$, and let $f : E \to \mathbb{R}^n$ be a function, and let $x_0 \in E$ be an interior point of E. Suppose $L_1, L_2 : \mathbb{R}^n \to \mathbb{R}^m$ are linear transformations s.t. f is differentiable at x_0 with derivative L_1 and also differentiable at x_0 with derivative L_2. Then $L_1 = L_2$.

Proof. Since x_0 is an interior point. By Remark 1.2.3, we have

$$\lim_{h \to 0} \frac{\|f(x_0 + h) - f(x_0) - L_1(h)\|}{\|h\|} = 0 \text{ and } \lim_{h \to 0} \frac{\|f(x_0 + h) - f(x_0) - L_2(h)\|}{\|h\|} = 0.$$

Thus,

$$\begin{aligned} 0 \leq \frac{\|L_1(h) - L_2(h)\|}{\|h\|} &= \frac{\|L_1(h) - f(x_0 + h) + f(x_0) + f(x_0 + h) - f(x_0) - L_2(h)\|}{\|h\|} \\ &\leq \frac{\|L_1(h) - f(x_0 + h) + f(x_0)\|}{\|h\|} + \frac{\|f(x_0 + h) + f(x_0) - L_2(h)\|}{\|h\|}. \end{aligned}$$

Thus, by squeeze theorem, we know

$$\lim_{h \to 0} \frac{\|L_1(h) - L_2(h)\|}{\|h\|} = 0.$$

Now take $v \in \mathbb{R}^n \setminus \{0\}$, and let $h = tv$, so

$$\lim_{t\to 0} \frac{\|L_1(tv) - L_2(tv)\|}{\|tv\|} = 0.$$

That is,

$$\lim_{t\to 0} \frac{|t|\,\|L_1(v) - L_2(v)\|}{|t|\|v\|} = 0.$$

Hence, we have

$$\lim_{t\to 0} \frac{\|L_1(v) - L_2(v)\|}{\|v\|} = 0,$$

and since $v \neq 0$, so we know $L_1(v) = L_2(v)$, and since v can be arbitrary non-zero vector in $\mathbb{R}^n \setminus \{0\}$, so $L_1 = L_2$. $(L_1(0) = L_2(0) = 0)$ ■

Remark 1.2.5. Because of Lemma 1.2.2, the derivative of f at interior points x_0 is unique, and thus we may safely write it as $f'(x_0)$ or $Df(x_0)$. Thus $f'(x_0)$ is the unique linear transformation $f'(x_0) : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{x\to x_0\ x\neq x_0} \frac{\|f(x) - (f(x_0) + f'(x_0)(x - x_0))\|}{\|x - x_0\|} = 0.$$

Informally, this condition means that near x_0, the function f can be approximated by its linearization:

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0).$$

This approximation is sometimes referred to as Newton's approximation in higher dimension.

Remark 1.2.6. A useful consequence of Lemma 1.2.2 is the following: If two functions f and g satisfy $f(x) = g(x)$ for all $x \in E$, and both are differentiable at an interior point x_0, then their derivatives coincide at x_0, i.e. $f'(x_0) = g'(x_0)$. This will be important in later arguments, where one extends functions to larger domains or modifies them on sets of measure zero.

Remark 1.2.7. As we have emphasized, Lemma 1.2.2 guarantees the uniqueness of the derivative only at interior points of the domain E. If x_0 is instead a boundary point of E, the derivative may fail to be uniquely determined.

1.3 Partial and Directional Derivatives

We now begin to relate the concept of differentiability to the more classical notions of partial and directional derivatives. Directional derivatives describe how f changes when we move from a point x_0 in a fixed direction v.

Definition 1.3.1 (Directional derivative)**.** Let $E \subseteq \mathbb{R}^n$, let $f : E \to \mathbb{R}^m$ be a function, let x_0 be an interior point of E and let $v \in \mathbb{R}^n$. If the limit

$$\lim_{\substack{t\to 0, t>0\\ x_0+tv\in E}} \frac{f(x_0 + tv) - f(x_0)}{t} = \lim_{\substack{t\to 0^+\\ x_0+tv\in E}} \frac{f(x_0 + tv) - f(x_0)}{t}$$

exists, then we say f is differentiable in the direction v at x_0. This limit is called the directional derivative of f at x_0 in the direction v, and we denote it by

$$D_v f(x_0) := \lim_{t\to 0^+} \frac{f(x_0 + tv) - f(x_0)}{t} \in \mathbb{R}^m.$$

Remark 1.3.1. Under this definition,

$$D_{-v}f(x_0) = \lim_{t\to 0^+} \frac{f(x_0 + t(-v)) - f(x_0)}{t} = \lim_{t\to 0^+} \frac{f(x_0 - tv) - f(x_0)}{t}.$$

In usual definition, one define

$$D_v f(x_0) = \lim_{t\to 0} \frac{f(x_0 + tv) - f(x_0)}{t}.$$

This usual definition implies that

$$D_v f(x_0) = -D_{-v}f(x_0)$$

since

$$\begin{aligned} D_v f(x_0) &= \lim_{t\to 0^+} \frac{f(x_0 + tv) - f(x_0)}{t} = \lim_{t\to 0^-} \frac{f(x_0 + tv) - f(x_0)}{t} \\ &= \lim_{s\to 0^+} = \frac{f(x_0 + (-s)v) - f(x_0)}{(-s)} = -\lim_{s\to 0^+} \frac{f(x_0 - sv) - f(x_0)}{s} = -(D_{-v}f(x_0)). \end{aligned}$$

Remark 1.3.2. This definition should be compared with the definition of differentiability. Here we divide by the scalar t rather than by a vector, so the expression always makes sense algebraically. The directional derivative $D_v f(x_0)$ is a vector in $\mathbb{R}^m$.

Remark 1.3.3. It is sometimes possible to define directional derivatives at boundary points of E, provided that the vector v points inward toward the domain. However, we shall restrict attention to interior points in what follows.

Lemma 1.3.1. Let $E \subseteq \mathbb{R}^n$, let $f : E \to \mathbb{R}^m$, let $x_0 \in \mathrm{Int}(E)$ and $v \in \mathbb{R}^n$. If f is differentiable at x_0, then f is differentiable in the direction v at x_0, and

$$D_v f(x_0) = f'(x_0)(v).$$

Proof. Since $x_0 \in \mathrm{Int}(E)$ and f is differentiable at x_0, so

$$\lim_{h\to 0} \frac{\|f(x_0 + h) - f(x_0) - f'(x_0)(h)\|}{\|h\|} = 0.$$

Note that for $t \neq 0$, we have

$$\frac{f(x_0 + tv) - f(x_0)}{t} = \frac{f(x_0 + tv) - f(x_0) - f'(x_0)(tv)}{t} + f'(x_0)(v),$$

so

$$\frac{f(x_0 + tv) - f(x_0)}{t} - f'(x_0)(v) = \frac{f(x_0 + tv) - f(x_0) - f'(x_0)(tv)}{t}.$$

Now we show that

$$\lim_{t\to 0^+} \frac{f(x_0 + tv) - f(x_0) - f'(x_0)(tv)}{t} = 0.$$

Recall that f is differentiable at x_0 and $x_0 \in \mathrm{Int}(E)$, so

$$\lim_{h\to 0} \frac{\|f(x_0 + h) - f(x_0) - f'(x_0)(h)\|}{\|h\|} = 0.$$

Let $h = tv$ when $v \neq 0$, then

$$\lim_{t\to 0^+} \frac{\|f(x_0 + tv) - f(x_0) - f'(x_0)(tv)\|}{\|tv\|} = 0 \Rightarrow \lim_{t\to 0^+} \frac{\|f(x_0 + tv) - f(x_0) - f'(x_0)(tv)\|}{|t|} = 0,$$

which is equivalent to

$$\lim_{t\to 0^+} \frac{f(x_0+tv)-f(x_0)-f'(x_0)(tv)}{t} = 0,$$

(since the numerator should tend to 0 so the limit will exist), so we know

$$\lim_{t\to 0^+} \frac{f(x_0+tv)-f(x_0)}{t} - f'(x_0)(v) = 0,$$

i.e.

$$\lim_{t\to 0^+} \frac{f(x_0+tv)-f(x_0)}{t} = f'(x_0)(v).$$

■

Remark 1.3.4. One consequence of this lemma is that total differentiability implies directional differentiability. However, the converse is not true: the existence of all directional derivatives does not guarantee differentiability.

Definition 1.3.2 (Partial Derivative)**.** Let $E \subseteq \mathbb{R}^n$, let $f : E \to \mathbb{R}^m$, and let $x_0 \in \operatorname{Int}(E)$, and let $1 \le j \le n$. The partial derivative of f with respect to the variable x_j at x_0, denoted $\frac{\partial f}{\partial x_j}(x_0)$, is defined by

$$\frac{\partial f}{\partial x_j}(x_0) := \lim_{t\to 0, t\neq 0} \frac{f(x_0+te_j)-f(x_0)}{t}$$

provided the limit exists.

Remark 1.3.5. If $f = \begin{pmatrix} f_1 \\ \vdots \\ f_m \end{pmatrix}$, then differentiation is componentwise:

$$\frac{\partial f}{\partial x_j}(x_0) = \begin{pmatrix} \frac{\partial f_1}{\partial x_j}(x_0) \\ \vdots \\ \frac{\partial f_m}{\partial x_j}(x_0) \end{pmatrix}.$$

We say that f is continuously differentiable if all partial derivatives exist and are continuous.

Remark 1.3.6. If f is differentiable at x_0, then

$$\frac{\partial f(x_0)}{\partial x_j} = D_{e_j} f(x_0) = f'(x_0)(e_j).$$

Remark 1.3.7. If f is differentiable at x_0, then

$$D_v f(x_0) = f'(x_0)(v) = f'(x_0)\left(\sum_{j=1}^n v_j e_j\right) = \sum_{j=1}^n v_j f'(x_0)(e_j) = \sum_{j=1}^n v_j \frac{\partial f(x_0)}{\partial x_j}.$$

Theorem 1.3.1. Let $E \subseteq \mathbb{R}^n$, let $f : E \to \mathbb{R}^m$, $F \subseteq E$, and let $x_0 \in \operatorname{Int}(F)$ (the partial derivative exists on a neighborhood of x_0). If all partial derivative $\frac{\partial f}{\partial x_j}$ exists on F and are continuous at x_0, then f is differentiable at x_0, and

$$f'(x_0)(v) = \sum_{j=1}^n v_j \frac{\partial f}{\partial x_j}(x_0)$$

for all $v \in \mathbb{R}^n$.

Proof. Define a linear map $L : \mathbb{R}^n \to \mathbb{R}^m$ by

$$L(v) := \sum_{j=1}^{n} v_j \frac{\partial f}{\partial x_j}(x_0).$$

We prove that f is differentiable at x_0 with derivative L. It suffices to show that for any $\varepsilon > 0$, $\exists \delta > 0$ s.t.

$$\|f(x) - f(x_0) - L(x - x_0)\| < \varepsilon \|x - x_0\| \text{ whenever } x \in F \text{ and } \|x - x_0\| < \delta$$

Since $x_0 \in \operatorname{Int}(F)$, $\exists r > 0$ s.t. $B(x_0, r) \subseteq F$. Because each partial derivative $\frac{\partial f_i}{\partial x_j}$ is continuous at x_0, for every pair (i, j) there exists $\delta_{ij} > 0$ s.t.

$$\left| \frac{\partial f_i}{\partial x_j}(x) - \frac{\partial f_i}{\partial x_j}(x_0) \right| < \frac{\varepsilon}{nm} \text{ whenever } \|x - x_0\| < \delta_{ij}.$$

Let $\delta := \min\{r, \delta_{ij} : 1 \le i \le m, 1 \le j \le n\}$. Now fix $x \in F$ with $\|x - x_0\| < \delta$ and write

$$x - x_0 = \sum_{j=1}^{n} v_j e_j, \text{ so } x = x_0 + \sum_{j=1}^{n} v_j e_j.$$

Then

$$\|x - x_0\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}, \quad |v_j| \le \|x - x_0\|,$$

Write $f = (f_1, f_2, \ldots, f_m)$. To estimate $f(x) - f(x_0)$, we vary the coordinates one at a time. Consider

$$\begin{aligned} x^{(0)} &= x_0, \\ x^{(j)} &= x_0 + \sum_{k=1}^{j} v_k e_k \text{ for } j = 1, 2, \ldots, n. \end{aligned}$$

Thus, $x^{(n)} = x$ and

$$f(x) - f(x_0) = \sum_{j=1}^{n} \left(f\left(x^{(j)}\right) - f\left(x^{(j-1)}\right) \right) = \sum_{j=1}^{n} \sum_{i=1}^{m} \left(f_i\left(x^{(j)}\right) - f_i\left(x^{(j-1)}\right) \right) e_i.$$

Fix j and consider the j-th increment. For each component f_i, define

$$\varphi_i^j(t) = f_i\left(x^{(j-1)} + t v_j e_j\right), \quad 0 \le t \le 1.$$

Then

$$f_i\left(x^{(j)}\right) - f_i\left(x^{(j-1)}\right) = \varphi_i^j(1) - \varphi_i^j(0).$$

Since f_i has continuous partial derivatives, φ_i^j is differentiable. By MVT, there exists $t_j \in (0, 1)$ s.t.

$$\varphi_i^j(1) - \varphi_i^j(0) = \varphi_i^{j\prime}(t_j).$$

By the chain rule (See Lemma 1.3.2),

$$\varphi_i^{j\prime}(t) = \nabla f_i\left(x^{(j-1)} + t v_j e_j\right) \cdot (v_j e_j) = \frac{\partial f_i}{\partial x_j}\left(x^{(j-1)} + t v_j e_j\right) v_j.$$

Hence,

$$f_i\left(x^{(j)}\right) - f_i\left(x^{(j-1)}\right) = \frac{\partial f_i}{\partial x_j}(\xi_j)\, v_j,$$

where

$$\xi_j = x^{(j-1)} + t_j v_j e_j.$$

Subtracting $\frac{\partial f_i}{\partial x_j}(x_0)v_j$, we obtain

$$\left| f_i\left(x^{(j)}\right) - f_i\left(x^{(j-1)}\right) - \frac{\partial f_i}{\partial x_j}(x_0)v_j \right| = \left| \frac{\partial f_i}{\partial x_j}(\xi_j) - \frac{\partial f_i}{\partial x_j}(x_0) \right| |v_j|.$$

Because $\|\xi_j - x_0\| < \delta$, continuity of partial derivatives yields,

$$\left| f_i\left(x^{(j)}\right) - f_i\left(x^{(j-1)}\right) - \frac{\partial f_i}{\partial x_j}(x_0)v_j \right| < \frac{\varepsilon}{nm}|v_j|.$$

Summing over $i = 1, 2, \ldots, m$ and using $\|u\| \leq \sum_i |u_i|$, we conclude

$$\left\| f\left(x^{(j)}\right) - f\left(x^{(j-1)}\right) - \frac{\partial f}{\partial x_j}(x_0)v_j \right\| = \left\| \sum_{i=1}^{m} \left(f_i\left(x^{(j)}\right) - f_i\left(x^{(j-1)}\right) - \frac{\partial f_i}{\partial x_j}(x_0)v_j \right) e_i \right\|$$
$$\leq \sum_{i=1}^{m} \left| f_i\left(x^{(j)}\right) - f_i\left(x^{(j-1)}\right) - \frac{\partial f_i}{\partial x_j}(x_0)v_j \right| \leq m\frac{\varepsilon}{nm}|v_j| = \frac{\varepsilon}{n}|v_j| \leq \frac{\varepsilon}{n}\|x - x_0\|.$$

Finally, summing over $j = 1, 2, \ldots, n$ and applying the triangle inequality,

$$\left\| f(x) - f(x_0) - \sum_{j=1}^{n} \frac{\partial f}{\partial x_j}(x_0)v_j \right\| = \left\| \sum_{j=1}^{n} \left(f\left(x^{(j)}\right) - f\left(x^{(j-1)}\right) - \frac{\partial f}{\partial x_j}(x_0)v_j \right) \right\|$$
$$\leq \sum_{j=1}^{n} \left\| \left(f\left(x^{(j)}\right) - f\left(x^{(j-1)}\right) - \frac{\partial f}{\partial x_j}(x_0)v_j \right) \right\| \leq \sum_{j=1}^{n} \frac{\varepsilon}{n}\|x - x_0\| = \varepsilon\|x - x_0\|.$$

Since $L(x - x_0) = \sum_{j=1}^{n} v_j \frac{\partial f}{\partial x_j}(x_0)$, so we have

$$\|f(x) - f(x_0) - L(x - x_0)\| \leq \varepsilon\|x - x_0\|,$$

and we're done. ■

Remark 1.3.8. From Theorem 1.3.1 and Lemma 1.3.1 we conclude the following important fact:

If the partial derivatives of a function $f : E \to \mathbb{R}^m$ exist and are continuous on a set $F \subseteq E$, then at every interior point x_0 of F all directional derivatives exist, and they are given by

$$D_{(v_1, v_2, \ldots, v_n)} f(x_0) = \sum_{j=1}^{n} v_j \frac{\partial f}{\partial x_j}(x_0).$$

- The scalar-valued case. If $f : E \to \mathbb{R}$ is real-valued, we define the gradient of f at x_0 to be the row vector
$$\nabla f(x_0) := \left(\frac{\partial f}{\partial x_1}(x_0), \ldots, \frac{\partial f}{\partial x_n}(x_0) \right).$$
Whenever x_0 lies in the interior of a region where the partial derivatives exist and are continuous, the directional derivative takes the familiar form
$$D_v f(x_0) = \nabla f(x_0) \cdot v.$$

- The vector-valued case. Now let $f : E \to \mathbb{R}^m$ be vector-valued, say
$$f = \begin{pmatrix} f_1 \\ \vdots \\ f_m \end{pmatrix}.$$

If x_0 lies in the interior of a region where all partial derivatives exist and are continuous, then

$$f'(x_0)(v_1, v_2, \ldots, v_n) = \sum_{j=1}^{n} v_j \frac{\partial f}{\partial x_j}(x_0).$$

Writing this componentwise,

$$f'(x_0)(v_1, \ldots, v_n) = \sum_{j=1}^{n} v_j \begin{pmatrix} \frac{\partial f_1}{\partial x_j}(x_0) \\ \vdots \\ \frac{\partial f_m}{\partial x_j}(x_0) \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^{n} v_j \frac{\partial f_1}{\partial x_j}(x_0) \\ \vdots \\ \sum_{j=1}^{n} v_j \frac{\partial f_m}{\partial x_j}(x_0) \end{pmatrix} = \begin{pmatrix} \nabla f_1(x_0) \cdot v \\ \vdots \\ \nabla f_m(x_0) \cdot v \end{pmatrix} = \begin{pmatrix} \nabla f_1(x_0) \\ \vdots \\ \nabla f_m(x_0) \end{pmatrix} v$$

- The derivative matrix (Jacobian matrix). We therefore define the derivative matrix (or Jacobian matrix) of f at x_0 by

$$Df(x_0) := \left(\frac{\partial f_i}{\partial x_j}(x_0) \right)_{\substack{1 \le i \le m \\ 1 \le j \le n}}.$$

Explicitly,

$$Df(x_0) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(x_0) & \cdots & \frac{\partial f_1}{\partial x_n}(x_0) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(x_0) & \cdots & \frac{\partial f_m}{\partial x_n}(x_0) \end{pmatrix}.$$

With this notation, the derivative acts by matrix multiplication:

$$D_v f(x_0) = f'(x_0)v = Df(x_0)v.$$

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Lemma 1.3.2. Let $E \subseteq \mathbb{R}^n$ and let $f = (f_1, f_2, \ldots, f_m) : E \to \mathbb{R}^m$. Fix $1 \le i \le m$ and $1 \le j \le n$. Let $z \in E$, $v \in \mathbb{R}$ and define $\phi_i^j(t) = f_i(z + tve_j)$ for $0 \le t \le 1$. Assume that for some $t_0 \in (0, 1)$, $\exists \rho > 0$ s.t.

$$z + (t_0 + s)ve_j \in E \quad \text{for all } |s| < \rho,$$

and that the partial derivative $\frac{\partial f_i}{\partial x_j}$ exists at $p = z + t_0 ve_j$. Then, ϕ_i^j is differentiable at t_0 and

$$\left(\phi_i^j\right)'(t_0) = \frac{\partial f_i(p)}{\partial x_j}(v).$$

Proof. Note that

$$\begin{aligned} \left(\phi_i^j\right)'(t_0) &= \lim_{h \to 0} \frac{\phi_i^j(t_0 + h) - \phi_i^j(t_0)}{h} = \lim_{h \to 0} \frac{f_i(z + (t_0 + h)ve_j) - f_i(z + t_0 ve_j)}{h} \\ &= \lim_{h \to 0} \frac{f_i(p + hve_j) - f_i(p)}{h}. \end{aligned}$$

Now if $v = 0$, then this limit is equal to 0. Now if $v \neq 0$, then

$$\lim_{h \to 0} \frac{f_i(p + hve_j) - f_i(p)}{h} = \lim_{h \to 0} \frac{f_i(p + hve_j) - f_i(p)}{vh} \cdot v = \lim_{s \to 0} \frac{f_i(p + se_j) - f_i(p)}{s} \cdot v = \frac{\partial f_i(p)}{\partial x} \cdot v.$$

■

Example 1.3.1. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $f(x, y) = (x^2 + xy, y^2)$, then

$$Df(x, y) = \begin{pmatrix} \frac{\partial}{\partial x}(x^2 + xy) & \frac{\partial}{\partial y}(x^2 + xy) \\ \frac{\partial}{\partial x}(y^2) & \frac{\partial}{\partial y}(y^2) \end{pmatrix} = \begin{pmatrix} 2x + y & x \\ 0 & 2y \end{pmatrix}.$$

Thus,

$$D_{(v,w)}f(x,y) = \begin{pmatrix} 2x+y & x \\ 0 & 2y \end{pmatrix} \begin{pmatrix} v \\ w \end{pmatrix} = \begin{pmatrix} (2x+y)v + xw \\ 2yw \end{pmatrix}.$$

1.4 Chain Rule

If $f : X \to Y$ and $g : Y \to Z$, then their composition $g \circ f : X \to Z$ is defined by $(g \circ f)(x) = g(f(x))$. We want to show that

$$D(g \circ f)(x) = (Dg(f(x))) \cdot (Df(x))$$

under suitable conditions.

Definition 1.4.1. Let $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ be square matrix. The trace of A, denoted $\mathrm{Tr}(A)$, is defined by $\mathrm{Tr}(A) = \sum_{i=1}^{n} a_{in}$. Let $M = (m_{ij})$ be an $r \times s$ matrix. We may regard M as an element of $\mathbb{R}^{r\times s}$. We can define its Frobenius norm by

$$\|M\|_F = \left(\sum_{i=1}^{r} \sum_{j=1}^{s} m_{ij}^2 \right)^{\frac{1}{2}},$$

which can be written as $\|M\|_F = (\mathrm{Tr}\,(MM^t))^{\frac{1}{2}}$.

Theorem 1.4.1. Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$. Define the Frobenius norm by

$$\|M\|_F = \sqrt{\mathrm{Tr}\,(MM^t)} = \left(\sum_{i=1}^{r} \sum_{j=1}^{s} m_{ij}^2 \right)^{\frac{1}{2}}$$

with $M \in \mathbb{R}^{r\times s}$. Then,

$$\|AB\|_F \le \|A\|_F \|B\|_F.$$

In particular, for any column vector $v \in \mathbb{R}^n$, we have

$$\|Av\| \le \|A\|_F \|v\|,$$

and the linear map $x \mapsto Ax$ is continuous on $\mathbb{R}^n$.

Proof. Note that

$$(AB)_{ik} = \sum_{j=1}^{n} a_{ij} b_{jk},$$

so

$$|(AB)_{ik}| = \left| \sum_{j=1}^{n} a_{ij} b_{jk} \right| \le \left(\sum_{j=1}^{n} a_{ij}^2 \right)^{\frac{1}{2}} \left(\sum_{j=1}^{n} b_{jk}^2 \right)^{\frac{1}{2}}.$$

Hence, we have

$$\|AB\|_F^2 = \sum_i \sum_k (AB)_{ik}^2 \le \sum_i \sum_k \left(\sum_{j=1}^{n} a_{ij}^2 \right) \left(\sum_{j=1}^{n} b_{jk}^2 \right) = \|A\|_F^2 \|B\|_F^2.$$

For the second part, since we have

$$\|Av\|_F \le \|A\|_F \|v\|_F,$$

and we know for column vector, Frobenius norm coincides with the Euclidean norm, so we're done.

Finally, for $x, x_0 \in \mathbb{R}^n$, we know $Ax - Ax_0 = A(x - x_0)$, and thus

$$0 \le \|Ax - Ax_0\| = \|A(x - x_0)\| \le \|A\|_F \|x - x_0\|,$$

and if $x \to x_0$, we know

$$\|A\|_F \|x - x_0\| \to 0$$

since $\|A\|_F$ is a constant, and thus $\|A(x - x_0)\|$ also converges to 0, which shows the map $x \mapsto Ax$ is linear. ■

Theorem 1.4.2. Let $E \subseteq \mathbb{R}^n$, let $f : E \to \mathbb{R}^m$, and let x_0 be an interior point of E. If f is differentiable at x_0, then f is continuous at x_0.

Proof. If f is differentiable at x_0, then there exists $L : \mathbb{R}^n \to \mathbb{R}^m$ s.t.

$$\lim_{x \to x_0} \frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = 0.$$

Thus,

$$\lim_{x \to x_0} \|f(x) - f(x_0) - L(x - x_0)\| = \lim_{x \to x_0} \frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} \lim_{x \to x_0} \|x - x_0\| = 0.$$

Hence, we know

$$\begin{aligned} \|f(x) - f(x_0)\| &= \|f(x) - f(x_0) - L(x - x_0) + L(x - x_0)\| \\ &\leq \|f(x) - f(x_0) - L(x - x_0)\| + \|L(x - x_0)\| \end{aligned}$$

and the former converges to 0 as $x \to x_0$, while the latter also converges to 0 as $x \to x_0$ since L is continuous. ■

Theorem 1.4.3 (Chain Rule)**.** Let $E \subseteq \mathbb{R}^n$, let $F \subseteq \mathbb{R}^m$, and let $f : E \to F$ and $g : F \to \mathbb{R}^p$. Let $x_0 \in \operatorname{Int}(E)$. Suppose f is differentiable at x_0, that $f(x_0)$ lies in the interior of F, and that g is differentiable at $f(x_0)$. Then, $g \circ f : E \to \mathbb{R}^p$ is differentiable at x_0, and

$$(g \circ f)'(x_0) = g'(f(x_0)) \circ f'(x_0)$$

Proof. Set $y_0 = f(x_0)$ and denote the derivatives by $A = f'(x_0) : \mathbb{R}^n \to \mathbb{R}^m$, $B : g'(f(x_0)) = g'(y_0) : \mathbb{R}^m \to \mathbb{R}^p$. Since g is differentiable at y_0, define $\eta : \mathbb{R}^m \to \mathbb{R}^p$ by

$$\eta(k) = \begin{cases} \dfrac{g(y_0 + k) - g(y_0) - B(k)}{\|k\|}, & \text{if } k \neq 0; \\ 0, & \text{if } k = 0. \end{cases}$$

Note that $\lim_{k \to 0} \eta(k) = 0$ since g is differentiable at y_0, so η is continuous at 0. Moreover, we have

$$g(y_0 + k) - g(y_0) - Bk = \eta(k)\|k\| \quad \text{true for all } k.$$

Since $x_0 \in \operatorname{Int}(E)$, $x_0 + h \in E$ for small enough $\|h\|$. Note that

$$\begin{aligned} (g \circ f)(x_0 + h) - g(f(x_0)) &= g(f(x_0 + h)) - g(f(x_0)) = g(f(x_0) + (f(x_0 + h) - f(x_0))) - g(f(x_0)) \\ &= B(f(x_0 + h) - f(x_0)) + \eta(f(x_0 + h) - f(x_0)) \cdot \|f(x_0 + h) - f(x_0)\|, \end{aligned}$$

so we have

$$(g \circ f)(x_0 + h) - g(f(x_0)) - BAh = B(f(x_0 + h) - f(x_0) - Ah) + \eta(f(x_0 + h) - f(x_0)) \cdot \|f(x_0 + h) - f(x_0)\|.$$

Now since

$$\lim_{h \to 0} \frac{\|f(x_0 + h) - f(x_0) - Ah\|}{\|h\|} = 0,$$

so

$$\frac{\|B(f(x_0 + h) - f(x_0) - Ah)\|}{\|h\|} \leq \frac{\|B\|_F \|f(x_0 + h) - f(x_0) - Ah\|}{\|h\|}$$

implies

$$\lim_{h\to 0}\frac{\|B(f(x_0+h)-f(x_0)-Ah)\|}{\|h\|}=0.$$

Now we claim that

$$\lim_{h\to 0}\frac{\eta\left(f(x_0+h)-f(x_0)\right)\cdot\|f(x_0+h)-f(x_0)\|}{\|h\|}=0.$$

Note that

$$\frac{\eta\left(f(x_0+h)-f(x_0)\right)\cdot\|f(x_0+h)-f(x_0)\|}{\|h\|}=\eta\left(f(x_0+h)-f(x_0)\right)\frac{\|f(x_0+h)-f(x_0)\|}{\|h\|}.$$

Now since η is continuous at 0, so

$$\lim_{h\to 0}\eta(f(x_0+h)-f(x_0))=\eta(0)=0.$$

Also,

$$\begin{aligned}\frac{\|f(x_0+h)-f(x_0)\|}{\|h\|}&=\frac{\|f(x_0+h)-f(x_0)-f'(x_0)(h)+f'(x_0)(h)\|}{\|h\|}\\&\le\frac{\|f(x_0+h)-f(x_0)-f'(x_0)(h)\|+\|f'(x_0)(h)\|}{\|h\|}\\&\le\frac{\|f(x_0+h)-f(x_0)-f'(x_0)(h)\|}{\|h\|}+\frac{\|h\|\|f'(x_0)\|_F}{\|h\|}\\&\to\|A\|_F\text{ as }h\to 0.\end{aligned}$$

Thus,

$$\lim_{h\to 0}\frac{\|(g\circ f)(x_0+h)-g(f(x_0))-BAh\|}{\|h\|}=0.$$

■

Corollary 1.4.1. We may express the chain rule in matrix form.

$$D(g\circ f)(x_0)=Dg(f(x_0))Df(x_0).$$

Remark 1.4.1. We prove the chain rule of linear transformations in the last theorem, and earlier we have found that in fact the linear transformation $f'(x_0)$ is the Jacobian matrix $Df(x_0)$, so we can write the matrix form of the chain rule, i.e. the chain rule on Jacobian matrix.

Example 1.4.1. Let $f,g:\mathbb{R}^n\to\mathbb{R}$ be differentiable functions. Recall that

$$\nabla f=\left(\frac{\partial f}{\partial x_1},\frac{\partial f}{\partial x_2},\dots,\frac{\partial f}{\partial x_n}\right),\quad\nabla g=\left(\frac{\partial g}{\partial x_1},\frac{\partial g}{\partial x_2},\dots,\frac{\partial g}{\partial x_n}\right),$$

and

$$\nabla f=\sum_{i=1}^n\frac{\partial f}{\partial x_i}e_i,$$

then we have

$$\nabla(fg)=\sum_{i=1}^n\frac{\partial}{\partial x_i}(fg)e_i=\sum_{i=1}^n\left(\frac{\partial f}{\partial x_i}g+f\frac{\partial g}{\partial x_i}\right)e_i=g\nabla f+f\nabla g$$

by ordinary product rule(前微後不微加上前不微後微).

Example 1.4.2. Consider $f : \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}$ defined by

$$f(A) = A^t A,$$

then $Df(A) =?$

Proof. We use two methods to solve this problem.

Method 1: Direct Expansion (Using the Definition) To find the derivative of f at A, we apply a small matrix perturbation $H \in \mathbb{R}^{n\times n}$ and expand $f(A+H)$:

$$f(A+H) = (A+H)^t(A+H) = \left(A^t + H^t\right)(A+H) = A^tA + A^tH + H^tA + H^tH.$$

Hence, we have

$$f(A+H) - f(A) = \left(A^tH + H^tA\right) + H^tH.$$

Hence, we may choose $L : \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}$ by

$$L(H) = A^tH + H^tA.$$

Thus, we have

$$\lim_{H\to 0} \frac{\|f(A+H) - f(A) - L(H)\|}{\|H\|} = \lim_{H\to 0} \frac{\|H^tH\|}{\|H\|} \le \lim_{H\to 0} \frac{\|H^t\|\,\|H\|}{\|H\|} = \lim_{H\to 0} \left\|H^t\right\| = 0.$$

Method 2: Decomposition and the Chain Rule We can deconstruct f into

$$A \overset{\Phi}{\longmapsto} (A^t, A) \overset{\Psi}{\longmapsto} A^tA$$

Note that

$$\Phi(A+h) - \Phi(A) - \left(H^t, H\right) = 0,$$

so

$$\lim_{\|H\|\to 0} \frac{\|\Phi(A+H) - \Phi(A) - (H^t, H)\|}{\|H\|} = 0.$$

Thus, $\Phi(A)'(H) = (H^t, H)$. Also, we can check $\Psi'(B,C)(K,L) = KC + BL$, i.e.

$$\lim_{(K,L)\to(0,0)} \frac{\|\Psi(B+K, C+L) - \Psi(B,C) - (KC+BL)\|}{\|(K,L)\|} = 0.$$

Note that

$$\Psi(B+K, C+L) - \Psi(B,C) - (KC+BL) = KL.$$

and

$$\|(K,L)\| = \sqrt{\|K\|^2 + \|L\|^2} \ge \frac{\|K\| + \|L\|}{\sqrt{2}}.$$

Thus,

$$\begin{aligned}\frac{\|\Psi(B+K, C+L) - \Psi(B,C) - (KC+BL)\|}{\|(K,L)\|} &\le \sqrt{2}\frac{\|KL\|}{\|K\| + \|L\|} \le \sqrt{2}\frac{\|K\|\|L\|}{\|K\| + \|L\|} \\ &\le \sqrt{2}\frac{\|K\|\|L\|}{\|L\|} = \sqrt{2}\|K\|.\end{aligned}$$

This shows

$$0 \leq \lim_{(K,L)\to(0,0)} \frac{\|\Psi(B+K,C+L)-\Psi(B,C)-(KC+BL)\|}{\|(K,L)\|} \leq \lim_{(K,L)\to(0,0)} \sqrt{2}\|K\| = 0.$$

By chain rule, we know

$$f'(A)(H) = \Psi'(\Phi(A))\,(\Phi'(A)(H)) = H^tA + A^tH.$$

⊛

Lecture 4

If $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear, then $T'(x) = T$ for every x. This is because for every x 5 March.

$$\lim_{h\to 0} \frac{\|T(x+h)-T(x)-T(h)\|}{\|h\|} = 0.$$

Hence, for any differentiable $f : E \to \mathbb{R}^n$, the composition Tf is differentiable and

$$(Tf)'(x_0) = T'(f(x_0)) \circ f'(x_0) = T \circ f'(x_0) = T(f'(x_0)).$$

This generalizes the familiar one-variable rule $(cf)' = cf'$.

Another useful special case of the chain rule is the following. If $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable and $x_j : \mathbb{R} \to \mathbb{R}$ are differentiable functions for $j = 1, 2, \ldots, n$, then

$$\frac{\mathrm{d}}{\mathrm{d}t} f(x_1(t), x_2(t), \ldots, x_n(t)) = \sum_{j=1}^{n} x_j'(t) \frac{\partial f}{\partial x_j}(x_1(t), x_2(t), \ldots, x_n(t))$$

since we can suppose $f = (f_1, f_2, \ldots, f_m)$ and $g(t) = (x_1(t), x_2(t), \ldots, x_n(t))^t$ and thus

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} f(x_1(t), x_2(t), \ldots, x_n(t)) &= \frac{\mathrm{d}}{\mathrm{d}t} f \circ g(t) = f'(g(t)) \circ g'(t) \\
&= \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(g(t)) & \frac{\partial f_1}{\partial x_2}(g(t)) & \cdots & \frac{\partial f_1}{\partial x_n}(g(t)) \\ & \vdots & & \\ \frac{\partial f_m}{\partial x_1}(g(t)) & \frac{\partial f_m}{\partial x_2}(g(t)) & \ldots & \frac{\partial f_m}{\partial x_n}(g(t)) \end{pmatrix} \begin{pmatrix} x_1'(t) \\ \vdots \\ x_n'(t) \end{pmatrix} \\
&= \begin{pmatrix} \sum_{i=1}^n \frac{\partial f_1}{\partial x_i}(g(t)) x_i'(t) \\ \vdots \\ \sum_{i=1}^n \frac{\partial f_m}{\partial x_i}(g(t)) x_i'(t) \end{pmatrix} = \sum_{i=1}^n x_i'(t) \begin{pmatrix} \frac{\partial f_1}{\partial x_i}(g(t)) \\ \vdots \\ \frac{\partial f_m}{\partial x_i}(g(t)) \end{pmatrix} \\
&= \sum_{i=1}^n x_i'(t) \frac{\partial f}{\partial x_i}(g(t)) = \sum_{i=1}^n x_i'(t) \frac{\partial f}{\partial x_i}(x_1(t), x_2(t), \ldots, x_n(t)).
\end{aligned}$$

The Mean Value Theorem for functions $f : \mathbb{R} \to \mathbb{R}$ asserts that

$$f(y) - f(x) = f'(z)(y - x)$$

where $z \in (x, y)$. We begin by extending this statement to real-valued functions defined on open subsets of $\mathbb{R}^n$.

Theorem 1.4.4 (Mean Value Theorem for Real-Valued Functions)**.** Let V be open in $\mathbb{R}^n$ and suppose that $f : V \to \mathbb{R}$ is differentiable on V. Given $a, x \in V$, write $L(x; a)$ for the (closed) line segment joining a to x, namely

$$L(x; a) \coloneqq \{a + t(x - a) : 0 \leq t \leq 1\}.$$

Assume that $L(x;a) \subseteq V$. Then there exists $c \in L(x;a)$ such that

$$f(x) - f(a) = \nabla f(c) \cdot (x - a).$$

Proof. Suppose $g(t) := a + t(x - a)$ for $t \in \mathbb{R}$. Then, $g : \mathbb{R} \to \mathbb{R}^n$ is differentiable and

$$Dg(t) = x - a \quad \forall t \in \mathbb{R}.$$

Since V is open and the compact segment $L(x;a) = \{g(t) : t \in [0,1]\}$ is contained in V, there exists $\delta > 0$ s.t.

$$g(t) \in V \quad \text{for all } t \in I_\delta = (-\delta, 1 + \delta).$$

Hence, $f \circ g : I_\delta \to \mathbb{R}$ is well-defined and differentiable. By the chain rule, for each $t \in I_\delta$ we have

$$D(f \circ g)(t) = Df(g(t))D(g(t)) = Df(g(t))(x - a).$$

Because f is real-valued, the differential $Df(y)$ may be identified with the gradient by

$$Df(y)(h) = \nabla f(y) \cdot h, y \in V, h \in \mathbb{R}^n.$$

Therefore the above equation becomes

$$(f \circ g)'(t) = \nabla f(g(t)) \cdot (x - a), t \in I_\delta.$$

Now apply the one-dimensional MVT to $f \circ g$ on $[0,1]$. There exists $t_0 \in (0,1)$ such that

$$(f \circ g)(1) - (f \circ g)(0) = (f \circ g)(0)'(t_0).$$

Since $(f \circ g)(1) = f(x)$ and $(f \circ g)(0) = f(a)$, we obtain

$$f(x) - f(a) = \nabla f(g(t_0)) \cdot (x - a).$$

We can set $c := g(t_0) \in L(x;a)$, and we're done. ■

The conclusion of Theorem 1.4.4 relies essentially on the fact that f is real-valued. The following remark shows that no analogous statement can hold in general for vector-valued functions, even when the domain is one-dimensional.

Remark 1.4.2. The function $f(t) = (\cos t, \sin t)$ is differentiable on $\mathbb{R}$ and satisfies $f(w\pi) = f(0)$, but there is no $c \in \mathbb{R}$ s.t. $Df(c) = (0,0)$.

Now we want to extend the MVT above to $f : \mathbb{R}^n \to \mathbb{R}^m$ with $m > 1$, one can recover a valid statement by testing against linear functionals on $\mathbb{R}^m$. Equivalently, we take the dot product with an arbitrary vector $a \in \mathbb{R}^m$ and apply the real-valued mean value theorem to the scalar function $t \mapsto a \cdot f(x + tu)$. This yields the following useful generalization.

In the following statement, we let

$$L(x;y) = \{tx + (1-t)y : 0 \le t \le 1\}.$$

Theorem 1.4.5 (Mean-Value Theorem)**.** Let S be an open subset of $\mathbb{R}^n$ and assume that $f : S \to \mathbb{R}^m$ is differentiable at each point of S. Let x and y be two points in S s.t. $L(x;y) \subseteq S$. Then for every vector $a \in \mathbb{R}^m$ there is a point $z \in L(x;y)$ s.t.

$$a \cdot \{f(y) - f(x)\} = a \cdot \{f'(z)(y - x)\}.$$

Proof. Let $u = y - x$. Since S is open and $L(x;y) \subseteq S$, there is a $\delta > 0$ s.t.

$$x + tu \in S \text{ for all real } t \in (-\delta, 1 + \delta).$$

Let a be a fixed vector in $\mathbb{R}^m$ and define the real-valued function F on $(-\delta, 1 + \delta)$ by

$$F(t) = a \cdot f(x + tu).$$

Then F is differentiable on $(-\delta, 1+\delta)$ since $F = T \circ f$ where $T(x) = a \cdot x$ is a linear map, and note that

$$F'(t) = a \cdot \{f'(x+tu)(u)\}.$$

By the usual MVT for real-valued functions, we have

$$F(1) - F(0) = F'(\theta) \text{ for some } \theta \text{ with } 0 < \theta < 1.$$

Now

$$F'(\theta) = a \cdot \{f'(x+\theta u)(u)\} = a \cdot \{f'(z)(y-x)\},$$

where $z = x + \theta u \in L(x; y)$. But

$$F(1) - F(0) = a \cdot \{f(y) - f(x)\},$$

so we're done. ■

Remark 1.4.3. If S is convex, then $L(x; y) \subseteq S$ for all $x, y \in S$, so Theorem 1.4.5 holds for all x and y ion S.

1.5 Double Derivatives and Clairaut's theorem

Definition 1.5.1. Let $E \subseteq \mathbb{R}^n$ be open and let $f : E \to \mathbb{R}^m$. We say f is twice continuously differentiable on E if

(1) f is continuously differentiable on E, i.e. $\frac{\partial f}{\partial x_i}$ exists and are continuous on E

(2) for each $j = 1, 2, \ldots, n$, the partial derivative $\frac{\partial f}{\partial x_j} : E \to \mathbb{R}^m$ is continuously differentiable on E.

Theorem 1.5.1 (Clairaut's theorem)**.** Let E be an open set in $\mathbb{R}^n$, and let $f : E \to \mathbb{R}^m$ be twice continuously differentiable on E. Then, for all $1 \le i, j \le n$ and all $x_0 \in E$

$$\frac{\partial^2 f(x_0)}{\partial x_i \partial x_j} = \frac{\partial^2 f(x_0)}{\partial x_j \partial x_i}.$$

Proof. Since $f = (f_1, f_2, \ldots, f_m)$ where $f_i : E \to \mathbb{R}$, we may just assume $f : E \to \mathbb{R}$. We may also assume $x_0 = 0$. The $i = j$ case is trivial. Given any $\varepsilon > 0$, $\exists \delta_0 > 0$ s.t.

$$\left| \frac{\partial^2 f(x)}{\partial x_i \partial x_j} - \frac{\partial^2 f(0)}{\partial x_i \partial x_j} \right| < \varepsilon \text{ and } \left| \frac{\partial^2 f(x)}{\partial x_j \partial x_i} - \frac{\partial^2 f(0)}{\partial x_j \partial x_i} \right| < \varepsilon$$

whenever $\|x\| \le 2\delta_0$ since the second order derivative is continuous at 0. Fix $\delta \in (0, \delta_0)$. Consider the square in the (x_i, x_j)-plane with vertices $0, \delta e_i, \delta e_j, \delta e_i + \delta e_j$ and define the second finite difference

$$X := f(\delta e_i + \delta e_j) - f(\delta e_i) - f(\delta e_j) + f(0).$$

Define

$$\phi(t) := f(te_i + \delta e_j) - f(te_i), \quad t \in [0, \delta].$$

Then, $X = \phi(\delta) - \phi(0)$. By MVT, $\exists c_i \in (0, \delta)$ s.t.

$$X = \delta \phi'(c_i) = \delta \left[\frac{\partial f}{\partial x_i}(c_i e_i + \delta e_j) - \frac{\partial f}{\partial x_i}(c_i e_i) \right].$$

Now consider

$$g(s) := \frac{\partial f}{\partial x_i}(c_i e_i + s e_j), \quad s \in [0, \delta].$$

Then

$$g(\delta) - g(0) = \frac{\partial f}{\partial x_i}(c_i e_i + \delta e_j) - \frac{\partial f}{\partial x_i}(c_i e_i).$$

Applying MVT one more time, then we know there exists $c_j \in (0, \delta)$ s.t.

$$g(\delta) - g(0) = g'(c_j)\delta = \delta \frac{\partial^2 f}{\partial x_j \partial x_i}(c_i e_i + c_j e_j).$$

Substituting into the expression for X gives

$$X = \delta^2 \frac{\partial^2 f}{\partial x_j x_i}(\xi), \quad \xi = c_i e_i + c_j e_j.$$

Since $\|\xi\| = \sqrt{c_i^2 + c_j^2} < \delta\sqrt{2} < 2\delta_0$, continuity implies

$$\left| \frac{X}{\delta^2} - \frac{\partial^2 f}{\partial x_j \partial x_i}(0) \right| \leq \varepsilon.$$

If we write X as

$$X = f(\delta e_i + \delta e_j) - f(\delta e_j) - f(\delta e_i) + f(0)$$

and define

$$\psi(s) := f(\delta e_i + s e_j) - f(s e_j), \quad s \in [0, \delta].$$

Then, we can use similar method to show

$$\left| \frac{X}{\delta^2} - \frac{\partial^2 f}{\partial x_i \partial x_j}(0) \right| \leq \varepsilon.$$

Thus, by triangle inequality, we know

$$\left| \frac{\partial^2 f}{\partial x_j \partial x_i}(0) - \frac{\partial^2 f}{\partial x_i \partial x_j}(0) \right| \leq 2\varepsilon$$

for any $\varepsilon > 0$, and we're done. ■

Lecture 5

1.6 The Contraction Mapping Theorem

10 March.

Definition 1.6.1. Let (X, d) be a metric, and let $f : X \to X$ be a map. We say f is a contraction if

$$d(f(x), f(y)) \leq d(x, y) \quad \text{for all } x, y \in X.$$

We say f is a strict contraction if there exists a constant $0 < c < 1$ s.t.

$$d(f(x), f(y)) \leq c d(x, y) \quad \text{for all } x, y \in X.$$

We call c a contraction constant of f.

Example 1.6.1. Suppose $f : \mathbb{R} \to \mathbb{R}$,

(a) if $f(x) = x + 1$, then

$$f(x) - f(y) = (x + 1) - (y + 1) = x - y \Rightarrow |f(x) - f(y)| = |x - y|,$$

so f is a contraction but not a strict contraction.

(b) if $f(x) = \frac{x}{2}$, then

$$|f(x) - f(y)| = \frac{|x-y|}{2},$$

so f is a strict contraction.

Example 1.6.2. Suppose $f : [0,1] \to [0,1]$ and $f(x) = x - x^2$, then $f'(x) = 1 - 2x$, so

$$|f'(x)| \leq 1 \quad \text{on } [0,1].$$

By MVT,

$$f(x) - f(y) = f'(c)(x-y) \text{ for all } x, y \in [0,1] \text{ for some } c \in (x,y).$$

Now since $|f'(c)| \leq 1$, so we know f is a contraction. But f is not a strict contraction. Suppose it is a strict contraction, then

$$|f(x)| = |f(x) - f(0)| \leq c|x| \text{ for some } 0 < c < 1 \text{ for all } x \in [0,1].$$

However,

$$\lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} = f'(0) = 1,$$

which is impossible.

Definition 1.6.2. Let $f : X \to X$ be a map and let $x \in X$. We say that x is a fixed point of f if $f(x) = x$.

Remark 1.6.1. Not every contraction has a fixed point.

Remark 1.6.2. Suppose $X \subseteq \mathbb{R}$. The existence of a fixed point x s.t. $f(x) = x$ means that the graph of $y = f(x)$ intersect with $y = x$.

Definition 1.6.3. A metric space (X, d) is said to be complete if every Cauchy sequence in X converges to a point of X.

Example 1.6.3. In $\mathbb{R}^n$, $\mathbb{R}^n$ is complete with the standard metric

$$d(x,y) = \|x - y\| = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2} \text{ with } x = (x_1, x_2, \ldots, x_n),\ y = (y_1, y_2, \ldots, y_n).$$

Remark 1.6.3. Recall that

- Any closed subset of a complete metric space is also a complete metric space.
- Any closed set in $\mathbb{R}^n$ is complete. In particular, a closed ball

$$\overline{B(x_0, r)} = \{y : \|y - x_0\| \leq r\}$$

is complete.

Theorem 1.6.1 (Contraction Mapping Theorem)**.** Let (X, d) be a metric space and $f : X \to X$ be a strict contractor. Then f have at most one fixed point. Moreover, if we also assume X is non-empty and complete, then f has exactly one fixed point.

Proof. Suppose x and y are fixed points of a strict contractor f. Then, $f(x) = x$ and $f(y) = y$, and $\exists 0 < c < 1$ s.t.

$$d(x, y) = d(f(x), f(y)) \leq cd(x, y),$$

so $0 \leq (1 - c)d(x, y) \leq 0$, which shows $x = y$.

Now we assume X is non-empty and complete. Choose any point $x_0 \in X$. Consider the sequence $x_1 = f(x_0), x_2 = f(x_1), \ldots$. For $n \geq 1$, we have

$$d(x_{n+1}, x_n) = d(f(x_n), f(x_{n-1})) \leq cd(x_n, x_{n-1}) \leq \cdots \leq c^n d(x_1, x_0).$$

This shows $\{x_n\}_{n=0}^{\infty}$ is a Cauchy sequence. Now since X is complete, so $\lim_{n\to\infty} x_n = x^*$ for some $x^* \in X$. Now we claim $f(x^*) = x^*$. By triangle inequality and $f(x_{n-1}) = x_n$,

$$\begin{aligned} d(x^*, f(x^*)) &\leq d(x^*, f(x_n)) + d(f(x_n), f(x^*)) \leq d(x^*, x_{n+1}) + cd(x_n, x^*) \\ &\leq d(x^*, x_{n+1}) + d(x_n, x^*) \leq d(x_{n+1}, x_n) \to 0. \end{aligned}$$

■

Lemma 1.6.1. Let $B(0, r)$ be an open ball in $\mathbb{R}^n$, and let $g : B(0, r) \to \mathbb{R}^n$ be a map with $g(0) = 0$ and

$$\|g(x) - g(y)\| \leq \frac{1}{2}\|x - y\| \text{ for all } x, y \in B(0, r).$$

Then, the function $f : B(0, r) \to \mathbb{R}^n$ defined by $f(x) = x + g(x)$ is one to one and the image of $f(B(0, r))$ contains the ball $B\left(0, \frac{r}{2}\right)$.

Proof. First, we show that f is one to one. Suppose $f(x) = f(y)$, then

$$x + g(x) = y + g(y) \Leftrightarrow x - y = g(y) - g(x) \Leftrightarrow \|x - y\| = \|g(x) - g(y)\| \leq \frac{1}{2}\|x - y\|,$$

so $x = y$.

Now we show $B\left(0, \frac{r}{2}\right) \subseteq f(B(0, r))$. It suffices to show that for any $y \in B\left(0, \frac{r}{2}\right)$, $\exists x \in B(0, r)$ s.t. $f(x) = y$, or equivalently, $x = y - g(x)$. Define $F(x) = y - g(x)$, then we want to show that for any $y \in B\left(0, \frac{r}{2}\right)$, $\exists x \in B(0, r)$ s.t. $F(x) = x$. Now $y \in B\left(0, \frac{r}{2}\right)$, then $\exists \varepsilon > 0$ s.t. $2\|y\| \leq r - \varepsilon$.

Claim 1.6.1. $F : \overline{B(0, r - \varepsilon)} \to \overline{B(0, r - \varepsilon)}$ is a strict contraction.

Proof. We know

$$\|F(x)\| \leq \|y\| + \|g(x)\| \leq \frac{r - \varepsilon}{2} + \|g(x)\|.$$

Since $g(0) = 0$, so

$$\|g(x)\| = \|g(x) - g(0)\| \leq \frac{1}{2}\|x - 0\| \leq \frac{r - \varepsilon}{2}.$$

Thus,

$$\|F(x)\| \leq \frac{r - \varepsilon}{2} + \frac{r - \varepsilon}{2} = r - \varepsilon.$$

Thus, $F(x) \in \overline{B(0, r - \varepsilon)}$. Also,

$$\|F(x_1) - F(x_2)\| = \|g(x_1) - g(x_2)\| \leq \frac{1}{2}\|x_1 - x_2\|.$$

Thus, F is a strict contraction. ⊛

Now since closed ball is non-empty and complete, so by Theorem 1.6.1, F has a fixed point. ■

Lecture 6

12 March.

1.7 Inverse Function Theorem

Recall that if $f : \mathbb{R} \to \mathbb{R}$ is invertible, i.e. $f^{-1} : \mathbb{R} \to \mathbb{R}$ exists s.t. $f \circ f^{-1}(x) = f^{-1} \circ f(x) = x$ for all $x \in \mathbb{R}$, and $f'(x_0) \neq 0$ for some $x_0 \in \mathbb{R}$, then f^{-1} is differentiable at $f(x_0)$ and

$$\left(f^{-1}\right)'(f(x_0)) = \frac{1}{f'(x_0)},$$

which can be proved by chain rule.

In fact, one can say something even when f' is not invertible, as long as we know that f is continuously differentiable. If $f'(x_0) \neq 0$, then $f'(x_0)$ must be either strictly positive or strictly negative (since we assume f' to be continuous by Theorem 1.4.2), which implies $f'(x)$ is strictly positive for x near x_0, or strictly negative for x near x_0, and thus one-to-one in such small interval. In either case, f will become invertible if we restrict the domain and codomain of f to be sufficiently close to x_0 and to $f(x_0)$, respectively.

Now if $f : \mathbb{R}^n \to \mathbb{R}^n$ and $f \in C^1$, i.e. f is continuously differentiable. Suppose

$$f'(x_0) = Df(x_0) = \left[\frac{\partial f_i(x_0)}{\partial x_j}\right]$$

is invertible. Then, $\exists$ a neighborhood U of x_0 and V of $f(x_0)$ s.t. $f : U \to V$ is bijective and its inverse $f^{-1} : V \to U$ is continuous and differentiable at $f(x_0)$ with

$$\left(f^{-1}\right)'(f(x_0)) = [f'(x_0)]^{-1}.$$

This is the Inverse Function Theorem, and we will prove it later.

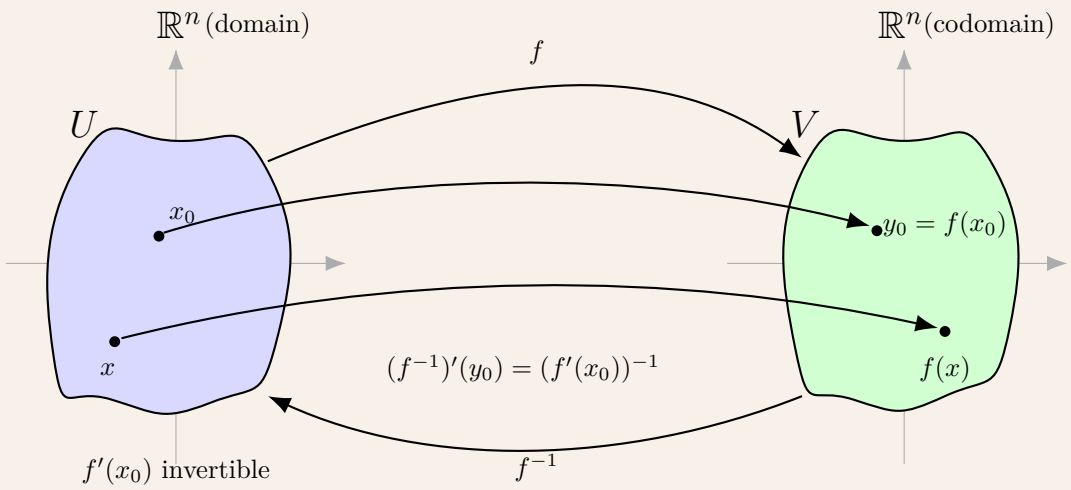
First, we prepare a lemma.

Lemma 1.7.1. Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be an invertible linear transformation . Then $T^{-1} : \mathbb{R}^n \to \mathbb{R}^n$ is also linear.

Proof. ■ DIY

Theorem 1.7.1 (The Inverse Function Theorem)**.** Let E be an open subset of $\mathbb{R}^n$ and let $f : E \to \mathbb{R}^n$ be a function which is continuously differentiable on E. Suppose $x_0 \in E$ is the point s.t. the linear map $f'(x_0) : \mathbb{R}^n \to \mathbb{R}^n$ is invertible, then there exists an open set $U \subseteq E$ s.t. $x_0 \in U$ and an open set $V \subseteq \mathbb{R}^n$ s.t. $f(x_0) \in V$ so that $f : U \to V$ is a bijection. In particular, there is an inverse map $f^{-1} : V \to U$. Furthermore, this inverse map is differentiable at $f(x_0)$, and

$$\left(f^{-1}\right)'(f(x_0)) = (f'(x_0))^{-1}.$$



Proof. Suppose such f^{-1} exists, i.e. $\exists U, V$ s.t. $f : U \to V$ and $f^{-1} : V \to U$ are bijective where $x_0 \in U$ and $f(x_0) \in V$. Since $f'(x_0)$ is invertible and $f^{-1}(f(x_0))$ exists, we know

$$f^{-1}(f(x)) = x \text{ for } x \in U \text{ and } \left(f^{-1}\right)'(f(x_0)) \circ (f'(x_0)) = Id \text{ by chain rule.}$$

Now we have to show such U, V exists. Let $A = f'(x_0)$, then

$$\lim_{x \to 0} \frac{\|f(x_0 + x) - f(x) - A(x)\|}{\|x\|} = 0.$$

Note that A^{-1} exists and

$$\begin{aligned} A^{-1}\left(f(x_0 + x) - f(x_0) - Ax\right) &= A^{-1}(f(x_0 + x) - f(x)) - A^{-1}Ax \\ &= A^{-1}\left(f(x_0 + x) - f(x_0)\right) - x. \end{aligned}$$

Note that

$$\lim_{x \to 0} \frac{\left\|A^{-1}(f(x_0 + x) - f(x_0)) - x\right\|}{\|x\|} = 0$$

since

$$\begin{aligned} 0 \le \lim_{x \to 0} \frac{\left\|A^{-1}(f(x_0 + x) - f(x_0)) - x\right\|}{\|x\|} &= \lim_{x \to 0} \frac{\left\|A^{-1}\left(f(x_0 + x) - f(x) - A(x)\right)\right\|}{\|x\|} \\ \le \lim_{x \to 0} \frac{\|A^{-1}\| \, \|f(x_0 + x) - f(x) - A(x)\|}{\|x\|} &= 0. \end{aligned}$$

Let $g(x) = A^{-1}(f(x_0 + x) - f(x_0)) - x$. Thus,

$$\lim_{x \to 0} \frac{\|g(x)\|}{\|x\|} = 0.$$

Note that $g(0) = 0$. Thus,

$$\lim_{x \to 0} \frac{\|g(x) - g(0) - 0\|}{\|x\|} = 0,$$

which shows g is differentiable at 0 with derivative $g'(0) = 0$. Now let's analyze $g(x)$. Since f is continuously differentiable, g is also continuously differentiable. Since E is open and $x_0 \in E$, there exists $r_0 > 0$ s.t. $B(x_0, r_0) \subseteq E$. Hence, g is defined on $B(0, r_0)$. (For $x \in B(0, r_0)$, $\|x\| < r_0$, so $\|(x + x_0) - x_0\| < r_0$, i.e. $x + x_0 \in B(x_0, r_0) \subseteq E$, so $g(x)$ is well-defined.)

Thus, for all i, j, $\exists r > 0$ s.t.

$$\left|\frac{\partial g_j}{\partial x_i}\right| \le \frac{1}{2\sqrt{n^5}} \text{ for } x \in B(0, r)$$

since $\frac{\partial g_j(0)}{\partial x_i} = 0$ (since $g'(0) = 0$ and thus g's Jacobian matrix at 0 is 0 matrix) and $\frac{\partial g_j}{\partial x_i}$ is continuous, and hence

$$\left|\frac{\partial g_j(x)}{\partial x_i} - \frac{\partial g_j(0)}{\partial x_i}\right| \le \frac{1}{2\sqrt{n^5}} \text{ if } \|x\| < r \text{ for some small enough } r.$$

By this, we can choose r small enough so that g is defined on $B(0, r)$ and

$$\left\|\frac{\partial g(x)}{\partial x_j}\right\| = \left(\sum_{i=1}^{n} \left(\frac{\partial g_i(x)}{\partial x_j}\right)^2\right)^{\frac{1}{2}} \le \frac{1}{2n^2} \text{ for all } x \in B(0, r), \quad j = 1, 2, \ldots$$

Now if $x, y \in B(0,r)$, then $x + t(y-x) \in B(0,r)$ if $0 \le t \le 1$. Also,

$$\begin{aligned} g(y) - g(x) &= \int_0^1 \left(\frac{\mathrm{d}}{\mathrm{d}t} \left[g(x + t(y-x)) \right] \right) \mathrm{d}t = \int_0^1 \left[Dg(x + t(y-x)) \circ (y-x) \right] \mathrm{d}t \\ &= \int_0^1 \left(D_{(y-x)} g(x + t(y-x)) \right) \mathrm{d}t. \end{aligned}$$

Now we estimate $D_v g(x)$ for $v = (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n$.

$$\|D_v g(x)\| = \left\| \sum_{j=1}^n v_j \frac{\partial g(x)}{\partial x_j} \right\| \le \sum_{j=1}^n \|v_j\| \left\| \frac{\partial g(x)}{\partial x_j} \right\| \le \sum_{j=1}^n \|v\| \frac{1}{2n^2} = \frac{\|v\|}{2n} \le \frac{\|v\|}{2}.$$

Thus,

$$\left\| D_{(y-x)} g(x + t(y-x)) \right\| \le \frac{\|x-y\|}{2}.$$

Hence, we have

$$\|g(y) - g(x)\| = \left\| \int_0^1 \left(D_{(y-x)} g(x + t(y-x)) \right) \mathrm{d}t \right\| \le \frac{\|x-y\|}{2}.$$

Recall $A^{-1}(f(x_0 + x) - f(x)) = g(x) + x$ and $g(0) = 0$, and we have shown

$$\|g(x) - g(y)\| \le \frac{\|x-y\|}{2} \text{ for } x, y \in B(0,r).$$

Let $\Phi(x) = x + g(x)$. By Lemma 1.6.1, $\Phi : B(0,r) \to \mathbb{R}^n$ is one to one and $B\left(0, \frac{r}{2}\right) \subseteq \Phi\left(B(0,r)\right)$. Since Φ is one to one on $B(0,r)$. Let $W = \Phi^{-1}\left(B\left(0, \frac{r}{2}\right)\right)$, then $\Phi : W \to B\left(0, \frac{r}{2}\right)$ is one to one and onto. Now we know Φ is continuous (since g is composition of differentiable and thus continuous map, so g is continuous, and thus Φ is continuous), so W is open (preimage of open set is open for continuous map). Also, $0 \in W$ since $\Phi(0) = 0 \in B\left(0, \frac{r}{2}\right)$. Now we claim $\Phi^{-1} : B\left(0, \frac{r}{2}\right) \to W$ is continuous. Note that for all $x, y \in W$

$$\|\Phi(x) - \Phi(y)\| = \|x - y + g(x) - g(y)\| \ge \|x-y\| - \|g(x) - g(y)\| \ge \|x-y\| - \frac{1}{2}\|x-y\| = \frac{\|x-y\|}{2}.$$

Let $x = \Phi^{-1}(u)$ and $y = \Phi^{-1}(v)$ for $u, v \in B\left(0, \frac{r}{2}\right)$, then

$$\left\| \Phi\left(\Phi^{-1}(u)\right) - \Phi\left(\Phi^{-1}(v)\right) \right\| \ge \frac{1}{2} \|\Phi^{-1}(u) - \Phi^{-1}(v)\|.$$

Thus,

$$\|u - v\| \ge \frac{1}{2} \left\| \Phi^{-1}(u) - \Phi^{-1}(v) \right\|.$$

This gives

$$\left\| \Phi^{-1}(u) - \Phi^{-1}(v) \right\| \le 2\|u - v\|,$$

and implies Φ^{-1} is continuous. Thus, $\Phi : W \to B\left(0, \frac{r}{2}\right)$ is a homomorphism, i.e. Φ is continuous and Φ^{-1} is also continuous. Since $\Phi(0) = 0$ and Φ is one-to-one, $\Phi^{-1}(0) = 0$. Also, note that

$$\lim_{z \to 0} \frac{\left\| \Phi^{-1}(z) - \Phi^{-1}(0) - z \right\|}{\|z\|} = \lim_{z \to 0} \frac{\|\Phi^{-1}(z) - z\|}{\|z\|}.$$

Note that for $z \ne 0$, if we let $\Phi^{-1}(z) = x$, then $z = \Phi(x) = x + g(x)$, and $x \ne 0$, so

$$\frac{\left\| \Phi^{-1}(z) - z \right\|}{\|z\|} = \frac{\|x - (x + g(x))\|}{\|z\|} = \frac{\|g(x)\|}{\|z\|} = \frac{\|g(x)\|}{\|x\|} \cdot \frac{\|x\|}{\|z\|}.$$

Hence,

$$\lim_{z\to 0}\frac{\left\|\Phi^{-1}(z)-z\right\|}{\|z\|}=\lim_{z\to 0}\frac{\|g(x)\|}{\|x\|}\cdot\frac{\|x\|}{\|z\|}=\lim_{z\to 0}\frac{\|g(x)\|}{\|x\|}\cdot\frac{\left\|\Phi^{-1}(z)\right\|}{\|z\|}$$

Now if $z\to 0$, then since Φ^{-1} is continuous, so $x=\Phi^{-1}(z)\to\Phi^{-1}(0)=0$. Also, when $z\to 0$, we know

$$\left\|\Phi^{-1}(z)-\Phi^{-1}(0)\right\|\le 2\|z-0\|\Rightarrow\frac{\left\|\Phi^{-1}(z)\right\|}{\|z\|}\le 2,$$

and since

$$\lim_{z\to 0}\frac{\|g(x)\|}{\|x\|}=\lim_{x\to 0}\frac{\|g(x)\|}{\|x\|}=0,$$

so by Squeeze theorem,

$$\lim_{z\to 0}\frac{\left\|\Phi^{-1}(z)-z\right\|}{\|z\|}=0.$$

This shows $\left(\Phi^{-1}\right)'(0)=I$. Recall that

$$\Phi(x)=x+g(x)=A^{-1}(f(x+x_0)-f(x_0))$$

and $\Phi:W\to B\left(0,\frac{r}{2}\right)$ is a homomorpism. Now set

$$U=x_0+W,\quad V=f(x_0)+A\left(B\left(0,\frac{r}{2}\right)\right).$$

Since W is open and A is a linear isomorphism, both U and V are open subsets of $\mathbb{R}^n$. Moreover, $x_0\in U$ and $f(x_0)\in V$. We claim that f maps U bijectively to V. Let $u=x+x_0\in U$, where $x\in W$, then since $\Phi(W)=B\left(0,\frac{r}{2}\right)$, we have $\Phi(x)\in B\left(0,\frac{r}{2}\right)$, and thus

$$f(u)=f(x+x_0)=f(x_0)+A\Phi(x)\in f(x_0)+A\left(B\left(0,\frac{r}{2}\right)\right)=V.$$

This shows $f(U)\subseteq V$. Conversely, let $y\in V$, then

$$A^{-1}(y-f(x_0))\in B\left(0,\frac{r}{2}\right).$$

Because $\Phi:W\to B\left(0,\frac{r}{2}\right)$ is surjective, there exists $x\in W$ s.t.

$$\Phi(x)=A^{-1}(y-f(x_0)).$$

For this x we have

$$f(x+x_0)=f(x_0)+A\Phi(x)=y.$$

This shows $V\subseteq f(U)$ and thus $f(U)=V$, i.e. $f:U\to V$ is surjective. Since Φ is one to one, so the x above is unique. It follows that f is one to one on U. Thus, $f:U\to V$ is bijective.

Now for $y\in V$ we know $y=f(x+x_0)$ for some $x\in W$, so $x+x_0=f^{-1}(y)$. Recall

$$\Phi(x)=A^{-1}(f(x+x_0)-f(x_0))=A^{-1}(y-f(x_0)).$$

Hence,

$$x=\Phi^{-1}\left(A^{-1}(y-f(x_0))\right)\Rightarrow f^{-1}(y)-x_0=\Phi^{-1}\left(A^{-1}(y-f(x_0))\right).$$

Thus,

$$f^{-1}(y)=\Phi^{-1}\left(A^{-1}(y-f(x_0))\right)+x_0.$$

This shows

$$\left(f^{-1}\right)'(f(x_0))=\left(\Phi^{-1}\right)'\left(A^{-1}(f(x_0)-f(x_0))\right)(A^{-1})'(0)=(\Phi^{-1})'(0)\circ A^{-1}=IA^{-1}=(f'(x_0))^{-1}$$

since $x_0\in U$ so $f(x_0)\in V$ and every linear transformation is differentiable at every point where $\left(A^{-1}\right)(x)=A^{-1}$ for all x. Thus, we're done. ■

Remark 1.7.1. Since $\det f'(x_0) \neq 0$ and det is continuous, there exists a neighborhood B of x_0 such that $f'(x)$ is invertible for all $x \in B$.

Applying the inverse function theorem at each point $x \in B$, we obtain neighborhoods U_x such that f^{-1} is differentiable on $f(U_x)$ and

$$(f^{-1})'(f(x)) = (f'(x))^{-1}.$$

Restrict to a sufficiently small neighborhood $U' \subset B$ of x_0 and set $V' = f(U')$. Then f^{-1} is differentiable on V'.

Moreover, for $y \in V'$,

$$(f^{-1})'(y) = (f'(f^{-1}(y)))^{-1}.$$

Since f^{-1} is continuous, f' is continuous, and matrix inversion is continuous on the set of invertible matrices, it follows that $(f^{-1})'$ is continuous on V'.

Hence, f^{-1} is continuously differentiable on V'.

Lecture 7

Remark 1.7.2. The requirement of $f'(x_0)$ is invertible is necessary to get the differentiability of f^{-1} at x_0. 17 March.

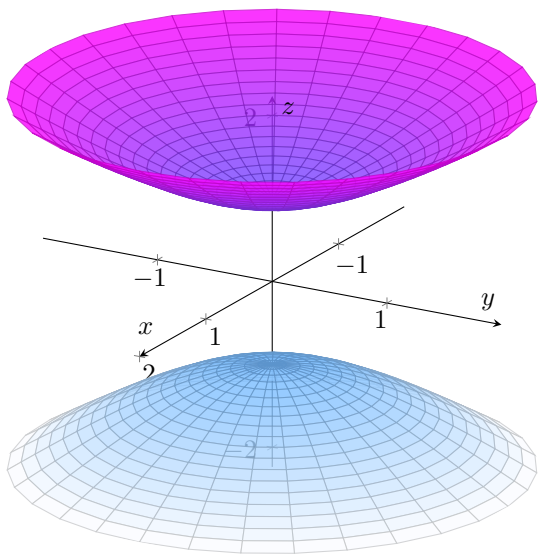
1.8 The Implicit Function Theorem

In $\mathbb{R}^2$, $S = \{(x, y) \mid x^2 + y^2 = 1\}$ is a circle. However, S is not the graph of one function. S can be expressed as the union of the graph of four functions:

$$\begin{cases} y = \sqrt{1 - x^2} \text{ for } -1 < x < 1 \\ y = -\sqrt{1 - x^2} \text{ for } -1 < x < 1 \\ x = \sqrt{1 - y^2} \text{ for } -1 < y < 1 \\ x = -\sqrt{1 - y^2} \text{ for } -1 < y < 1. \end{cases}$$

For a $\mathbb{R}^3$ case, consider $S = \{(x, y, z) \mid x^2 + y^2 - z^2 + 1 = 0\}$, then S is the union of 2 graph:

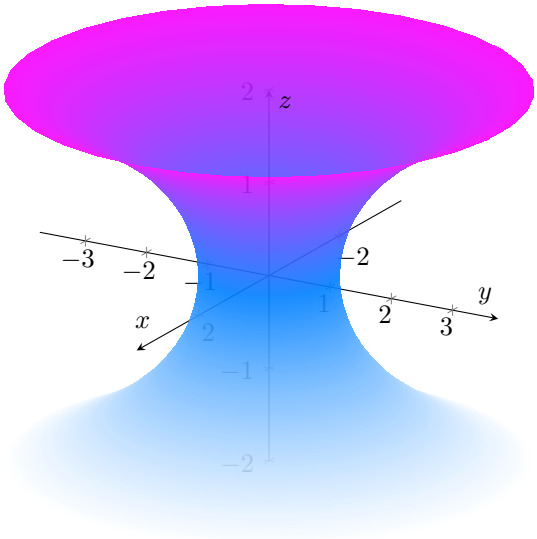
$$\begin{cases} z = \sqrt{x^2 + y^2 + 1} \text{ for } z > 0 \\ z = -\sqrt{x^2 + y^2 + 1} \text{ for } z < 0. \end{cases}$$



For another example, consider $S = \{(x, y, z) \mid x^2 + y^2 = \cosh^2(z)\}$, where

$$\cosh(z) = \frac{e^z + e^{-z}}{2}.$$

Then S is symmetric with respect to z.



Theorem 1.8.1 (Implicit Function Theorem). Let E be an open subset of $\mathbb{R}^n$, let $f : E \to \mathbb{R}$ be continuously differentiable, and let $y = (y_1, y_2, \ldots, y_n) \in E$ s.t. $f(y) = 0$ and $\frac{\partial f(y)}{\partial x_n} \neq 0$. Then, $\exists$ an open subset $U \subseteq \mathbb{R}^{n-1}$ containing $(y_1, y_2, \ldots, y_{n-1})$, an open subset $V \subseteq E$ containing y, and a function $g : U \to \mathbb{R}$ s.t.

$$g(y_1, y_2, \ldots, y_{n-1}) = y_n$$

and

$$\{x = (x_1, \ldots, x_n) \in V : f(x) = 0\} = \{(x_1, \ldots, x_{n-1}, g(x_1, \ldots, x_{n-1})) : (x_1, \ldots, x_{n-1}) \in U\}.$$

In other words, the set $\{x \in V : f(x) = 0\}$ is a graph of a function over U. Moreover, g is differentiable at $(y_1, \ldots, y_{n-1})$ and we have

$$\frac{\partial g}{\partial x_j}(y_1, y_2, \ldots, y_{n-1}) = -\frac{\frac{\partial f}{\partial x_j}(y)}{\frac{\partial f}{\partial x_n}(y)}$$

for all $1 \leq j \leq n - 1$.

Proof. Define $F : E \to \mathbb{R}^n$ by

$$(x_1, x_2, \ldots, x_n) \to (x_1, x_2, \ldots, x_{n-1}, f(x_1, x_2, \ldots, x_n)).$$

Since $f \in C^1(E)$, so $F \in C^1(E)$. Thus,

$$DF(x) = \begin{bmatrix} \nabla x_1 \\ \nabla x_2 \\ \vdots \\ \nabla x_{n-1} \\ \nabla f \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ & & \ddots & \\ 0 & \cdots & 1 & 0 \\ \frac{\partial f}{\partial x_1}(x) & \cdots & \frac{\partial f}{\partial x_{n-1}}(x) & \frac{\partial f}{\partial x_n}(x) \end{bmatrix},$$

which gives

$$\det(DF(x)) = \frac{\partial f(x)}{\partial x_n}.$$

Hence,

$$\det(DF(y)) = \frac{\partial f(y)}{\partial x_n} \neq 0,$$

so $F'(y)$ is invertible. Now we can apply inverse function theorem, so $\exists$ an open set $V \subseteq E$ and $W \subseteq \mathbb{R}^m$ with $y \in V$ and $F(y) = (y_1, \ldots, y_{n-1}, 0) \in W$ s.t. $F : V \to W$ is a C^1 diffeomorphism. Write the inverse in coordinate

$$F^{-1}(\underbrace{u_1, u_2, \ldots, u_n}_{u}) = (h_1(u), h_2(u), \ldots, h_n(u)).$$

Since $F(F^{-1}(u)) = u$ for $u \in W$, so

$$(h_1(u), \ldots h_{n-1}(u), f(h_1(u), \ldots, h_n(u))) = F(h_1(u), h_2(u), \ldots, h_n(u)) = (u_1, u_2, \ldots, u_n).$$

Hence,

$$\begin{aligned} h_1(u) &= u_1 \\ h_2(u) &= u_2 \\ &\vdots \\ h_{n-1}(u) &= u_{n-1} \\ f(h_1(u), \ldots, h_n(u)) &= f(u_1, u_2, \ldots, u_{n-1}, h_n(u)) = u_n. \end{aligned}$$

Thus, for $u_1, \ldots, u_n \in W$,

$$F^{-1}(u_1, \ldots, u_n) = (u_1, \ldots, u_{n-1}, h_n(u)).$$

By this, we can plug $u_n = 0$ into it and get

$$F^{-1}(u_1, \ldots, u_{n-1}, 0) = (u_1, \ldots, u_{n-1}, h_n(u_1, \ldots, u_{n-1}, 0)),$$

where

$$0 = u_n = f(u_1, \ldots, u_{n-1}, h_n(u_1, \ldots, u_{n-1}, 0)).$$

Now we want to restrict $u_n = 0$ and define g. Set

$$U = \left\{x' = (x_1, \ldots, x_{n-1}) \in \mathbb{R}^{n-1}, (x', 0) \in W\right\}.$$

Then U is open and contains $(y_1, \ldots, y_{n-1})$ since

$$F(y_1, \ldots, y_n) = (y_1, \ldots, y_{n-1}, f(y_1, \ldots, y_n)) = (y_1, \ldots, y_{n-1}, 0) \in W.$$

Define $g : U \to \mathbb{R}$ by

$$g(x') = g(x_1, \ldots, x_{n-1}) = h_n(x_1, \ldots, x_{n-1}, 0).$$

Since $f(y_1, \ldots, y_n) = 0$, we know $F(y) = (y_1, \ldots, y_n, 0)$. Hence,

$$(y_1, \ldots, y_n) = y = F^{-1}(F(y)) = F^{-1}(y_1, \ldots, y_n, 0) = (y_1, \ldots, y_{n-1}, h_n(y_1, \ldots, y_{n-1}, 0)).$$

Thus,

$$y_n = h_n(y_1, \ldots, y_{n-1}, 0) = g(\underbrace{y_1, \ldots, y_{n-1}}_{y'}) \Rightarrow g(y') = y_n.$$

Moreover, g is differentiable since h_n is differentiable.

Now we want to show the zero set is the graph of g.

Claim 1.8.1. $\{x \in V : f(x) = 0\} = \{(x', g(x')) : x' \in U\}$.

Proof. Let $A = \{x \in V : f(x) = 0\}$ and $B = \{(x', g(x')) : x' \in U\}$. We first show $A \subseteq B$. Let $x = (x', x_n) \in V$ and $f(x', x_n) = 0$. Then

$$F(x) = (x', f(x', x_n)) = (x', 0) \in W.$$

Since $x' \in U$,

$$x = F^{-1}(F(x)) = F^{-1}(x', 0) = (x', h_n(x', 0)) = (x', g(x')).$$

Hence, $A \subseteq B$.

Now we show $B \subseteq A$. Suppose $x' \in U$, then

$$x = (x', g(x')) = (x', h_n(x', 0)).$$

Thus,

$$F^{-1}(x', 0) = (x', h_n(x', 0)) = (x', g(x')).$$

Thus,

$$(x', 0) = F(F^{-1}(x', 0)) = F(x', g(x')) = (x', f(x', g(x'))).$$

Thus,

$$f(x', g(x')) = 0,$$

which shows $x \in A$. ⊛

Now for $(x_1, x_2, \ldots, x_{n-1}) \in U$, we know

$$f(x_1, x_2, \ldots, x_{n-1}, g(x_1, x_2, \ldots, x_{n-1})) = 0.$$

Fix $j \in [n-1]$, and set $y' = (y_1, \ldots, y_{n-1})$. Consider

$$\psi(t) \coloneqq f(y' + te_j, g(y' + te_j)),$$

defined for t near 0. Note that $\psi(t) \equiv 0$ for all t sufficiently close to 0. Hence, $\psi'(0) = 0$. Since f is differentiable at $y = (y', g(y'))$ and g is differentiable at y', we may apply the chain rule to compute $\psi'(0)$. Suppose $\psi(t) = f(\gamma(t))$, where $\gamma : \mathbb{R} \to \mathbb{R}^n$ is defined by

$$t \mapsto (y' + te_j, g(y' + te_j)).$$

Now since

$$\psi'(t) = f'(\gamma(t)) \circ \gamma'(t),$$

and f is a function from a subset of $\mathbb{R}^n$ to $\mathbb{R}$, so

$$f'(\gamma(t)) = \nabla f(\gamma(t)) = \left(\frac{\partial f(\gamma(t))}{\partial x_1}, \frac{\partial f(\gamma(t))}{\partial x_2}, \ldots, \frac{\partial f(\gamma(t))}{\partial x_n} \right),$$

which gives

$$f'(\gamma(0)) = \left(\frac{\partial f(y)}{\partial x_1}, \frac{\partial f(y)}{\partial x_2}, \ldots, \frac{\partial f(y)}{\partial x_n} \right).$$

Also,

$$\gamma'(t) = \left(0, \ldots, 1, \ldots, 0, \frac{\mathrm{d}g(y' + te_j)}{\mathrm{d}t} \right) \Rightarrow \gamma'(0) = \left(0, \ldots, 1, \ldots, 0, \frac{\mathrm{d}g(y')}{\mathrm{d}t} \right),$$

and note that

$$\frac{\mathrm{d}g(y' + te_j)}{\mathrm{d}t} = \nabla g(y' + te_j) \circ e_j \Rightarrow \frac{\mathrm{d}g(y')}{\mathrm{d}t} = \nabla g(y') \circ e_j = \frac{\partial g(y')}{\partial x_j}.$$

Hence,

$$\begin{aligned}\psi'(0) &= \left(\frac{\partial f(y)}{\partial x_1}, \frac{\partial f(y)}{\partial x_2}, \ldots, \frac{\partial f(y)}{\partial x_n}\right) \circ \left(0, \ldots, 1, \ldots, 0, \frac{\partial g(y')}{\partial t}\right) \\ &= \frac{\partial f}{\partial x_j}(y) + \frac{\partial f}{\partial x_n}(y)\frac{\partial g}{\partial x_j}(y'),\end{aligned}$$

and thus

$$0 = \psi'(0) = \frac{\partial f}{\partial x_j}(y) + \frac{\partial f}{\partial x_n}(y)\frac{\partial g}{\partial x_j}(y').$$

Now since $\frac{\partial f}{\partial x_n}(y) \neq 0$, we have

$$\frac{\partial g}{\partial x_j}(y') = -\frac{\frac{\partial f}{\partial x_j}(y)}{\frac{\partial f}{\partial x_n}(y)}.$$

■

Example 1.8.1. Let $S = \left\{(x, y, z) \in \mathbb{R}^3 \mid xy + yz + xz = -1\right\}$, which we rewrite as

$$\left\{(x, y, z) \in \mathbb{R}^3 : f(x, y, z) = 0\right\},$$

where $f : \mathbb{R}^3 \to \mathbb{R}$ is the function

$$f(x, y, z) = xy + yz + xz + 1.$$

Clearly f is continuously differentiable, and $\frac{\partial f}{\partial z} = y + x$. Thus, for any $(x_0, y_0, z_0) \in S$ sith $y_0 + x_0 \neq 0$, one can write this surface (near (x_0, y_0, z_0)) as a graph of the form

$$\{(x, y, g(x, y)) : (x, y) \in U\}$$

for some open set U containing (x_0, y_0), and some function g which is differentiable at (x_0, y_0). Indeed one can implicitly obtain

$$\frac{\partial g}{\partial x}(x_0, y_0) = -\frac{y_0 + z_0}{y_0 + x_0}, \quad \text{and} \quad \frac{\partial g}{\partial y}(x_0, y_0) = -\frac{x_0 + z_0}{y_0 + x_0}.$$

Chapter 2

Lebesgue measure

Lecture 8

To overcome the limitations of the Riemann integral, particularly its inability to handle pointwise limits of functions, this chapter constructs a more robust theory of integration starting with the concept of measure. 19 March.

2.1 The goal: Lebsgue measure

Our goal is to define a concept of measurable set, which will be a special kind of subset of $\mathbb{R}^n$. For every such measurable set $\Omega \subseteq \mathbb{R}^n$, we'll define the Lebesgue measure $m(\Omega)$ to be a number in $[0, \infty]$.

Definition 2.1.1 (Measurable set). The concept of measurable set will obey the following properties:

(i) **(Borel property)** Every open set in $\mathbb{R}^n$ is measurable, as is every closed set.

(ii) **(Complementarity)** If Ω is measurable, then $\mathbb{R}^n \setminus \Omega$ is also measurable.

(iii) **(Boolean algebra propery)** If $(\Omega_j)_{j\in J}$ is any finite collection of measurable sets, then $\bigcup_{j\in J} \Omega_j$ and $\bigcap_{j\in J} \Omega_j$ are also measurable.

(iv) **(σ-algebra property)** If $(\Omega_j)_{j\in J}$ are any countable collection of measurable sets (so J is countable), then $\bigcup_{j\in J} \Omega_j$ and $\bigcap_{j\in J} \Omega_j$ are also measurable.

Note that some of these properties are redundant; for instance, (iv) will imply (iii), and once one knows all open sets are measurable, (ii) will imply that all closed sets are measurable also. The properties (i)-(iv) will ensure that virtually every set one cares about is measurable; though there do exist non-measurable sets.

Definition 2.1.2 (Lebsegue measure). To every measurable set Ω, we associate the Lebsegue measure $m(\Omega)$ of Ω, which will obey the following properties:

(v) **(Empty set)** The empty set $\varnothing$ has measure $m(\varnothing) = 0$.

(vi) **(Positivity)** We have $0 \leq m(\Omega) \leq +\infty$ for every measurable set Ω.

(vii) **(Monotocity)** If $A \subseteq B$, and A and B are both measurable, then $m(A) \leq m(B)$.

(viii) **(Finite sub-additivity)** If $(A_j)_{j\in J}$ are a finite collection of measurable sets, then

$$m\left(\bigcup_{j\in J} A_j\right) \leq \sum_{j\in J} m(A_j).$$

(ix) **(Finite additivity)** If $(A_j)_{j\in J}$ are a finite collection of *disjoint* measurable sets, then

$$m\left(\bigcup_{j\in J} A_j\right) = \sum_{j\in J} m(A_j).$$

(x) **(Countable sub-additivity)** If $(A_j)_{j\in J}$ are a countable collection of measurable sets, then

$$m\left(\bigcup_{j\in J} A_j\right) \leq \sum_{j\in J} m(A_j).$$

(xi) **(Countable additivity)** If $(A_j)_{j\in J}$ are a countable collection of *disjoint* measurable sets, then

$$m\left(\bigcup_{j\in J} A_j\right) = \sum_{j\in J} m(A_j).$$

(xii) **(Normalization)** The unit cube

$$[0,1]^n = \{(x_1,\ldots,x_n) \in \mathbb{R}^n : 0 \leq x_j \leq 1 \text{ for all } 1 \leq j \leq n\}$$

has measure $m([0,1]^n) = 1$.

(xiii) **(Translation invariance)** If Ω is a measurable set, and $x \in \mathbb{R}^n$, then $x+\Omega := \{x+y : y \in \Omega\}$ is also measurable, and $m(x+\Omega) = m(\Omega)$.

Again, many of these properties are redundant; for instance, the countable additivity property can be used to deduce the finite additivity property, which in turn can be used to derive monotonicity (when combined with the positivity property). One can also obtain the sub-additivity properties from the additivity ones. Note that $m(\Omega)$ can be $+\infty$, and so in particular some of the sums in the above properties may also equal $+\infty$; in this chapter we adopt the convention that an infinite sum $\sum_{j\in J} a_j$ of non-negative quantities a_j is equal to $+\infty$ if the sum is not absolutely convergent. (Since everything is non0negative we will never have to deal with indeterminate forms such as $(-\infty)+(+\infty)$.)

Our goal for this chapter can then be stated:

Theorem 2.1.1 (Existence of Lebsegue measure)**.** There exists a concept of a measurable set and a way to assign a number $m(\Omega)$ to every measurable set $\Omega \subseteq \mathbb{R}^n$, which obeys all of the properties (i) to (xiii).

Lecture 9

2.2 First attempt: Outer Measure

24 March.

Definition 2.2.1. An open box B in $\mathbb{R}^n$ is any set of the form

$$B = \prod_{i=1}^{n}(a_i,b_i) := \{(x_1,\ldots,x_n) \in \mathbb{R}^n : x_i \in (a_i,b_i) \text{ for all } 1 \leq i \leq n\},$$

where $b_i \geq a_i$ are real numbers. We define the volume $\mathrm{vol}(B)$ of this box to be the number

$$\mathrm{vol}(B) := \prod_{i=1}^{n}(b_i - a_i).$$

Definition 2.2.2. Let $\Omega \subseteq \mathbb{R}^n$ be a subset of $\mathbb{R}^n$. We say a collection $(B_j)_{j\in J}$ of boxes cover Ω iff $\Omega \subseteq \bigcup_{j\in J} B_j$.

Remark 2.2.1. An index set J is countable if it is finite or $\exists$ a one-to-one map $f : J \to \mathbb{N}$ where $\mathbb{N} = \{1, 2, \dots\}$ is the set of natural numbers.

Remark 2.2.2. The set of rational numbers is countable. We usually denote it by

$$\mathbb{Q} = \left\{ \frac{p}{q} \mid q \neq 0, p, q \in \mathbb{Z} \right\}.$$

Definition 2.2.3 (Outer measure)**.** If Ω is a set in $\mathbb{R}^n$, we define the outer measure $m^*(\Omega)$ of Ω to be the quantity

$$m^*(\Omega) := \inf \left\{ \sum_{j \in J} \operatorname{vol}(B_j) : (B_j)_{j \in J} \text{ is a collection of boxes covers } \Omega \text{ and } J \text{ at most countable.} \right\}.$$

Note that $m^*(\Omega)$ may be ∞. Also, since $\operatorname{vol}(B_j) \geq 0$, so $m^*(\Omega) \geq 0$.

Lemma 2.2.1. The space $\mathbb{R}^n$ can be covered by countably many translates of the open cube $\left(-\frac{1}{2}, \frac{1}{2}\right)^n$.

Proof. For each $m \in \mathbb{Z}^n$, define

$$Q_m := \frac{m}{2} + \left(-\frac{1}{2}, \frac{1}{2}\right)^n.$$

We claim $\mathbb{R}^n = \bigcup_{m \in \mathbb{Z}^n} Q_m$. Let $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$. For each i, set $m_i := \lfloor 2x_i \rfloor \in \mathbb{Z}$, then

$$\frac{m_i}{2} \leq x_i < \frac{1}{2} + \frac{m_i}{2} \Rightarrow x \in \frac{m}{2} + \left(-\frac{1}{2}, \frac{1}{2}\right)^n.$$

Note that $\mathbb{Z}^n$ is countable since $\mathbb{Z}$ is countable, and thus we're done. ■

Lemma 2.2.2 (Properties of outer measure)**.** Outer measure has the following 6 properties:

(v) **(Empty set)** The empty set $\varnothing$ has outer measure 0, i.e. $m^*(\varnothing) = 0$.

(vi) **(Positivity)** We have $0 \leq m^*(\Omega) \leq \infty$ for all measurable set Ω.

(vii) **(Monotocity)** If $A \subseteq B \subseteq \mathbb{R}^n$, then $m^*(A) \leq m^*(B)$.

(viii) **(Finite sub-additivity)** If $(A_j)_{j \in J}$ are finite collection of subsets of $\mathbb{R}^n$, then

$$m^*\left(\bigcup_{j \in J} A_j\right) \leq \sum_{j \in J} m^*(A_j).$$

(x) **(Countable sub-additivity)** If $(A_j)_{j \in J}$ are countable collection of subsets of $\mathbb{R}^n$, then

$$m^*\left(\bigcup_{j \in J} A_j\right) \leq \sum_{j \in J} m^*(A_j).$$

(xiii) **(Translation invariance)** If Ω is a subset of $\mathbb{R}^n$, and $x \in \mathbb{R}^n$, then $m^*(x + \Omega) = m^*(\Omega)$.

proof of (v). From the definition we know $m^*(\Omega) \geq 0$ for all sets $\Omega \subseteq \mathbb{R}^n$. Hence, $m^*(\varnothing) \geq 0$. Now given $\varepsilon > 0$, we know $\left(0, \varepsilon^{\frac{1}{n}}\right)^n$ is an open box that covers $\varnothing$. Thus,

$$m^*(\varnothing) \leq \operatorname{vol}\left(\left(0, \varepsilon^{\frac{1}{n}}\right)^n\right) = \varepsilon.$$

Thus, $m^*(\Omega) = 0$. ■

proof of (vi). It follows from the definition of m^*. ■

proof of (vii). If $m^*(B) = \infty$, then this is trivial. Now suppose $m^*(B) < \infty$. Recall

$$m^*(B) = \inf\left\{\sum_{j\in J} \mathrm{vol}(B_j) : (B_j)_{j\in J} \text{ are open boxes cover } B\right\}.$$

Also, for any countable collection of boxes covering B, say $(B_j)_{j\in J}$, it also covers A, so

$$\left\{\sum_{j\in J} \mathrm{vol}(B_j) : (B_j)_{j\in J} \text{ are open boxes cover } B\right\} \subseteq \left\{\sum_{j\in J} \mathrm{vol}(A_j) : (A_j)_{j\in J} \text{ are open boxes cover } A\right\},$$

which shows

$$m^*(A) \leq m^*(B).$$

■

proof of (x). If one of $m^*(A_j) = \infty$, then

$$\infty = m^*(A_j) \leq m^*\left(\bigcup_{j\in J} A_j\right),$$

so we're done. Now we assume $m^*(A_j) < \infty$ for all $j \in J$. Given $\varepsilon > 0$, $\exists$ open boxes $B_{j,k}$ s.t. $A_j \subseteq \bigcup_{k=1}^{\infty} B_{j,k}$ and

$$\sum_{k=1}^{\infty} \mathrm{vol}(B_{j,k}) \leq m^*(A_j) + \frac{\varepsilon}{2^j}$$

since the definition of m^* is infimum. Now since J is countable, so we can re-label it to $\mathbb{N}$. Hence,

$$\bigcup_{j=1}^{\infty} A_j \subseteq \bigcup_{j=1}^{\infty}\bigcup_{k=1}^{\infty} B_{j,k},$$

and note that R.H.S. is the union of a collection of boxes, so $\bigcup_{j=1}^{\infty} A_j$ is covered by them. This shows

$$m^*\left(\bigcup_{j=1}^{\infty} A_j\right) \leq \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} \mathrm{vol}(B_{j,k}) \leq \sum_{j=1}^{\infty} m^*(A_j) + \frac{\varepsilon}{2^j} = \sum_{j=1}^{\infty} m^*(A_j) + \varepsilon.$$

Now since this is true for all $\varepsilon > 0$, so

$$m^*\left(\bigcup_{j=1}^{\infty} A_j\right) \leq \sum_{j=1}^{\infty} m^*(A_j).$$

■

Proposition 2.2.1 (Outer measure of closed box)**.** For any closed box

$$B = \prod_{i=1}^{n} [a_i, b_i] = \{(x_1, x_2, \ldots, x_n) \mid a_i \leq x_i \leq b_i \text{ for } 1 \leq i \leq n\},$$

we have

$$m^*(B) = \prod_{i=1}^{n} (b_i - a_i).$$

Proof. Write

$$\operatorname{vol}(B) = \prod_{i=1}^{n}(b_i - a_i).$$

We first show that

$$m^*(B) \leq \operatorname{vol}(B).$$

Let $\varepsilon > 0$, then the box

$$U_\varepsilon \coloneqq \prod_{i=1}^{n}(a_i - \varepsilon, b_i + \varepsilon)$$

contains B. Thus,

$$m^*(B) \leq \operatorname{vol}(U_\varepsilon) = \prod_{i=1}^{n}(b_i - a_i + 2\varepsilon).$$

Take $\varepsilon \to 0$, then we can show

$$m^*(B) \leq \operatorname{vol}(B).$$

Now we show

$$m^*(B) \geq \prod_{i=1}^{n}(b_i - a_i).$$

Claim 2.2.1. For any countable family of open boxes $\{B_j\}_{j\in J}$ covering B. We have

$$\operatorname{vol}(B) \leq \sum_{j\in J} \operatorname{vol}(B_j).$$

Proof. Since B is compact, $\exists$ a finite subcover of $(B_j)_{j\in J}$

$$(U_j)_{j=1}^{N} \text{ s.t. } B \subseteq \bigcup_{j=1}^{N} U_j.$$

Thus,

$$\sum_{j=1}^{N} \operatorname{vol}(U_j) \leq \sum_{j\in J} \operatorname{vol}(B_j).$$

Choose a closed box Q s.t. $B \subseteq Q$ and $U_k \subseteq Q$ for $1 \leq k \leq N$. Define functions on Q by

$$f(x) = \begin{cases} 1, & \text{if } x \in B; \\ 0, & \text{if } x \in Q \setminus B. \end{cases} \qquad f_k(x) = \begin{cases} 1, & \text{if } x \in U_k; \\ 0, & \text{if } x \in Q \setminus U_k. \end{cases}$$

Thus,

$$\int_Q f(x)\,\mathrm{d}x = \operatorname{vol}(B) \text{ and } \int_Q f_k(x)\,\mathrm{d}x = \operatorname{vol}(U_k).$$

Note that this is integrable since $B, (U_k)_{k=1}^{N}$ are all boxes and Q is also a box. Now since

$$f(x) \leq \sum_{j=1}^{k} f_j(x) \text{ on } Q.$$

(If $x \in B$, then $x \in U_j$ for some j.) Thus,

$$\operatorname{vol}(B) = \int_Q f(x)\,\mathrm{d}x \leq \int_Q \sum_{j=1}^{k} f_j(x)\,\mathrm{d}x = \sum_{j=1}^{k} \int_Q f_j(x)\,\mathrm{d}x = \sum_{j=1}^{k} \operatorname{vol}(U_j) \leq \sum_{j\in J} \operatorname{vol}(B_j).$$

⊛

This implies $\mathrm{vol}(B)$ is a lower bound of

$$\left\{\sum_{j\in J}\mathrm{vol}(B_j) : (B_j)_{j\in J} \text{ are boxes covering } B\right\},$$

so by the definition of m^*,

$$m^*(B) \geq \mathrm{vol}(B).$$

■

Lecture 10

26 March.

Corollary 2.2.1. For any open box

$$B = \prod_{i=1}^{n}(a_i, b_i) = \{(x_1, \ldots, x_n) \in \mathbb{R}^n \mid a_i < x_i < b_i \text{ for } 1 \leq i \leq n\},$$

we have $m^*(B) = \prod_{i=1}^{n}(b_i - a_i)$.

Proof. First since B cover itself, we have $m^*(B) \leq \mathrm{vol}(B) = \prod_{i=1}^{n}(b_i - a_i)$. Choose $\varepsilon_i < \frac{b_i - a_i}{2}$. Then,

$$[a_i + \varepsilon_i, b_i - \varepsilon_i] \subseteq (a_i, b_i).$$

Hence,

$$\prod_{i=1}^{n}[a_i + \varepsilon_i, b_i - \varepsilon_i] \subseteq \prod_{i=1}^{n}(a_i, b_i),$$

which shows

$$m^*\left(\prod_{i=1}^{n}[a_i + \varepsilon_i, b_i - \varepsilon_i]\right) \leq m^*(B),$$

and by previous proposition, we have

$$\prod_{i=1}^{n}(b_i - a_i - 2\varepsilon_i) = m^*\left(\prod_{i=1}^{n}[a_i + \varepsilon_i, b_i - \varepsilon_i]\right) \leq m^*(B).$$

Now pick all $\varepsilon_i \to 0$, then we have $m^*(B) \geq \mathrm{vol}(B)$. ■

Example 2.2.1. Let $r > 0$, then $(-r, r) \subseteq \mathbb{R}$, and thus

$$2r = m^*((-r, r)) \leq m^*(\mathbb{R})$$

for all r, so $m^*(\mathbb{R}) = \infty$. Similarly, $m^*(\mathbb{R}^n) = \infty$.

Example 2.2.2.

$$\mathbb{Q} = \text{ the set of rational numbers } = \bigcup_{q\in\mathbb{Q}}\{q\},$$

and $\mathbb{Q}$ is countable, so

$$m^*(\mathbb{Q}) \leq \sum_{q\in\mathbb{Q}} m^*(\{q\}) = 0.$$

In particular, $\exists$ an countable family of open boxes (B_j) s.t. $\mathbb{Q} \subseteq \bigcup_{j\in J} B_j$ and

$$\sum_{j\in J}\mathrm{vol}(B_j) \leq m^*(\mathbb{Q}) + \varepsilon = \varepsilon.$$

Example 2.2.3. Since $\mathbb{R} = (\mathbb{R} \setminus \mathbb{Q}) \cup \mathbb{Q}$, we have

$$m^*(\mathbb{R}) \leq m^*(\mathbb{R} \setminus \mathbb{Q}) + m^*(\mathbb{Q}),$$

and since $m^*(\mathbb{R}) = \infty$ and $m^*(\mathbb{Q}) = 0$, we know $m^*(\mathbb{R} \setminus \mathbb{Q}) = \infty$.

Claim 2.2.2. $m^*([0,1] \setminus \mathbb{Q}) = 1$.

Proof. Since

$$[0,1] = ([0,1] \setminus \mathbb{Q}) \cup ([0,1] \cap \mathbb{Q}),$$

we have

$$1 = m^*([0,1]) \leq m^*([0,1] \setminus \mathbb{Q}) + m^*([0,1] \cap \mathbb{Q}) = m^*([0,1] \setminus \mathbb{Q})$$

since

$$[0,1] \cap \mathbb{Q} \subseteq \mathbb{Q} \text{ and } 0 \leq m^*([0,1] \cap \mathbb{Q}) \leq m^*(\mathbb{Q}) = 0.$$

Now since

$$[0,1] \setminus \mathbb{Q} \subseteq [0,1],$$

so

$$m^*([0,1] \setminus \mathbb{Q}) \leq m^*([0,1]) = 1.$$

Thus,

$$m^*([0,1] \setminus \mathbb{Q}) = 1.$$

⊛

Claim 2.2.3. Let

$$L = \{(x,0) \mid x \in \mathbb{R}\} \subseteq \mathbb{R}^2,$$

then $m^*(L) = 0$.

Proof. Let $L_n = \{(x,0) \mid -n \leq x \leq n\} \subseteq \mathbb{R}^2$, then

$$L_n \subseteq [-n,n] \times [-\varepsilon,\varepsilon],$$

so

$$0 \leq m^*(L_n) \leq (2n) \cdot (2\varepsilon).$$

Let $\varepsilon \to 0$, then we have

$$m^*(L_n) = 0.$$

Now since

$$L = \bigcup_{n=1}^{\infty} L_n,$$

so

$$m^*(L) \leq \sum_{n=1}^{\infty} m^*(L_n) = 0,$$

which shows $m^*(L) = 0$. ⊛

2.3 Outer measure is not additive

We only know that

$$m^*\left(\bigcup_{j \in J} A_j\right) \leq \sum_{j \in J} m^*(A_j)$$

if J is countable. To construct a measure, we require that if $A_i \cap A_j = \varnothing$ for all $i, j \in J$ and $i \neq j$,

$$m^*\left(\bigcup_{j\in J} A_j\right) = \sum_{j\in J} m^*(A_j).$$

Proposition 2.3.1 (Failure of countable additivity)**.** There exists a countable collection $(A_j)_{j\in J}$ of disjoint subsets of $\mathbb{R}$ s.t.

$$m^*\left(\bigcup_{j\in J} A_j\right) \neq \sum_{j\in J} m^*(A_j).$$

Proof. We shall construct such a family from the coset of $\mathbb{Q}$ in $\mathbb{R}$.

- Step 1. Coset of $\mathbb{Q}$. For $x \in \mathbb{R}$, let

  $$x + \mathbb{Q} = \{x + q : q \in \mathbb{Q}\}.$$

  This is called a coset of $\mathbb{Q}$. Two properties of cosets:

  (1) $(x + \mathbb{Q}) \cap (y + \mathbb{Q}) \neq \varnothing$ implies $x + \mathbb{Q} = y + \mathbb{Q}$.

  (2) Every coset meets the interval $[0, 1]$. (We can pick $-x \leq q \leq 1 - x$ so that $0 \leq q + x \leq 1$)

- Step 2. Choose one representative from each coset. Let $\mathbb{R}/\mathbb{Q}$ denote the collection of all cosets. For each coset $A \in \mathbb{R}/\mathbb{Q}$. By (2) in Step 1, there exists $x_A \in [0, 1]$ s.t. $x_A \in A \cap [0, 1]$(By axiom of choice), we can choose one and define

  $$E = \{x_A : A \in \mathbb{R}/\mathbb{Q}, x_A \in [0, 1]\} \subseteq [0, 1].$$

- Step 3. Rational translates of E. Consider the set

  $$X := \bigcup_{q\in\mathbb{Q}\cap[-1,1]} (q + E).$$

  Since $E \subseteq [0, 1]$ and $-1 \leq q \leq 1$, we have $X \subseteq [-1, 2]$. Now we claim $[0, 1] \subseteq X$. Let y be any number in $[0, 1]$. Let $y \in [0, 1]$, then y is in some coset of $\mathbb{R}/\mathbb{Q}$. Thus, $\exists$ cosets A and $x_A \in E \subseteq [0, 1]$ s.t. $y \in A = x_A + \mathbb{Q} = y + \mathbb{Q}$. Let $y = x_A + q$ for some $q \in \mathbb{Q}$. Thus, $q = y - x_A \in [-1, 1]$. Thus, $q \in \mathbb{Q} \cap [-1, 1]$. Hence,

  $$y = q + x_A \in q + E \subseteq X.$$

  Thus,

  $$[0, 1] \subseteq X \subseteq [-1, 2].$$

- Step 4. the sets $q + E$ are pairwise disjoint. We now show that the family

  $$\{q + E\}_{q\in\mathbb{Q}\cap[-1,1]}$$

  is pairwise disjoint. Suppose, for contradiction, there are distinct $q, q' \in \mathbb{Q} \cap [-1, 1]$ we had

  $$(q + E) \cap (q' + E) \neq \varnothing.$$

  Then, there exists cosets $A, A' \in \mathbb{R}/\mathbb{Q}$ s.t.

  $$q + x_A = q' + x_{A'}.$$

Hence,

$$x_{A'} - x_A = q - q' \in \mathbb{Q}.$$

Now since $x_A \in A = x_A + \mathbb{Q}$ and $x_{A'} \in A' = x_{A'} + \mathbb{Q}$, so $A = A'$. But E contains exactly one representative from each coset, which means $x_A = x_{A'}$. Thus, $q = q'$, which contradicts to the assumption, so the sets $q + E$ are pairwise disjoint.

- Step 5. Comparing the two sides. Now since $[0,1] \subseteq X \subseteq [-1,2]$,

$$1 = m^*([0,1]) \leq m^*(X) \leq m^*([-1,2]) = 3.$$

On the other hand,

$$m^*(q+E) = m^*(E) \text{ for every } q \in \mathbb{Q} \cap [-1,1].$$

Thus,

$$\sum_{q \in \mathbb{Q} \cap [-1,1]} m^*(q+E) = \sum_{q \in \mathbb{Q} \cap [-1,1]} m^*(E).$$

Note that

$$\sum_{q \in \mathbb{Q} \cap [-1,1]} m^*(E) = \begin{cases} 0, & \text{if } m^*(E) = 0; \\ \infty, & \text{if } m^*(E) > 0. \end{cases}$$

In particular, it cannot lie in the interval $[1,3]$. But $m^*(X) \in [1,3]$. Hence,

$$m^*(X) \neq \sum_{q \in \mathbb{Q} \cap [-1,1]} m^*(q+E).$$

Since $\mathbb{Q} \cap [-1,1]$ is countably infinite, so we're done.

■

Proposition 2.3.2. There exists a finite collection $(A_j)_{j \in J}$ of disjoint subsets of $\mathbb{R}$ s.t.

$$m^*\left(\bigcup_{j \in J} A_j\right) \neq \sum_{j \in J} m^*(A_j).$$

Proof. Suppose m^* was finitely additive, and let E and X to be same as previous proposition, then

$$m^*(X) \leq \sum_{q \in \mathbb{Q} \cap [-1,1]} m^*(q+E) = \sum_{q \in \mathbb{Q} \cap [-1,1]} m^*(E).$$

Since we know that $1 \leq m^*(X) \leq 3$, we thus have $m^*(E) \neq 0$, or otherwise we will have $m^*(X) \leq 0$, but we have $m^*(X) \in [1,3]$.

Since $m^*(E) > 0$, so $m^*(E) > \frac{1}{n}$ for some $n \in \mathbb{N}$. Now let J be a finite subset of $\mathbb{Q} \cap [-1,1]$ of cardinality $3n$. If m^* were finitely additive, then we would have

$$m^*\left(\bigcup_{j \in J} (q+E)\right) = \sum_{q \in J} m^*(q+E) = \sum_{q \in J} m^*(E) > 3n \cdot \frac{1}{n} = 3.$$

But we know $\bigcup_{j \in J}(q+E)$ is a subset of X, so it has outer measure at most 3, so we have a contradiction. ■

Lecture 11

31 March.

2.4 Measurable sets

In this chapter, our goal is to construct a measure. Note that we constructed outer measure in previous lecture, but countbale additivity fails on it.

Definition 2.4.1 (Lebesgue measurability). Let $E \subseteq \mathbb{R}^n$. We say E is Lebesgue measurable, or measurable, iff we have the identity

$$m^*(A) = m^*(A \cap E) + m^*(A \setminus E) \text{ for all } A \subseteq \mathbb{R}^n.$$

Notation. If E is measurable, then we define $m(E) = m^*(E)$.

Remark 2.4.1. We know $A = (A \cap E) \cup (A \setminus E)$, so

$$m^*(A) \leq m^*(A \cap E) + m^*(A \setminus E).$$

Thus, E is measurable iff

$$m^*(A) \geq m^*(A \cap E) + m^*(A \setminus E).$$

Note that $\mathbb{R}^n$ is measurable since $A \cap \mathbb{R}^n = A$ and $A \setminus \mathbb{R}^n = \varnothing$. Also, $\varnothing$ is measurable since $A \cap \varnothing = \varnothing$ and $A \setminus \varnothing = A$.

Exercise 2.4.1. If A is an open box in $\mathbb{R}^n$ and $E = \{(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n : x_n > 0\}$, then

$$m^*(A) = m^*(A \cap E) + m^*(A \setminus E).$$

Proof. We first give a claim:

Claim 2.4.1. If A is an open interval in $\mathbb{R}$, then

$$m^*(A) = m^*(A \cap (0, \infty)) + m^*(A \setminus (0, \infty)).$$

Proof. Suppose $A = (a_i, b_i)$ s.t. $a_i \leq b_i$, then

- Case 1: $0 \leq a_i \leq b_i$, then $A \cap (0, \infty) = A$ and $A \setminus (0, \infty) = \varnothing$, so this is true.
- Case 2: $a_i \leq 0 \leq b_i$. Similar.
- Case 3: $a_i \leq b_i \leq 0$. Similar.

⊛

Suppose $A = \prod_{i=1}^n I_i$. Note that

$$m^*(A) = \text{vol}(A) = \prod_{i=1}^n \text{vol}(I_i) = \prod_{i=1}^n m_1^*(I_i).$$

Also,

$$A \cap E = \left(\prod_{i=1}^{n-1} I_i\right) \times (I_n \cap (0, \infty)) \quad A \setminus E = \left(\prod_{i=1}^{n-1} I_i\right) \times (I_n \setminus (0, \infty)).$$

Thus, $A \cap E$ and $A \setminus E$ are both open boxes. By the claim, we have

$$m_1^*(I_n) = m_1^*(I_n \cap (0, \infty)) + m_1^*(I_n \setminus (0, \infty)).$$

Thus,

$$\begin{aligned} m^*(A) &= \prod_{i=1}^{n} m_1^*(I_i) = \prod_{i=1}^{n-1} m_1^*(I_i) \times m_1^*(I_n) = \prod_{i=1}^{n-1} m_1^*(I_i) \times (m_1^*(I_n \cap (0,\infty)) + m_1^*(I_n \setminus (0,\infty))) \\ &= \prod_{i=1}^{n-1} m_1^*(I_i) \times m_1^*(I_n \cap (0,\infty)) + \prod_{i=1}^{n-1} m_1^*(I_i) \times m_1^*(I_n \setminus (0,\infty)) \\ &= m^*(A \cap E) + m^*(A \setminus E). \end{aligned}$$

■

Lemma 2.4.1 (Half-spaces are measurable). The half space $\{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_n > 0\}$ is measurable.

Proof. Let $E = \{x \in \mathbb{R}^n \mid x_n > 0\}$. Then

$$m^*(A) \leq m^*(A \cap E) + m^*(A \setminus E)$$

is true for all $A \subseteq \mathbb{R}^n$, so it suffices to show that

$$m^*(A) \geq m^*(A \cap E) + m^*(A \setminus E) \text{ for all } A \subseteq \mathbb{R}^n.$$

If $m^*(A) = \infty$, then this is obviously true. Now we assume $m^*(A) < \infty$. Given $\varepsilon > 0$, $\exists$ a countable collection of open boxes (B_k) s.t. $A \subseteq \bigcup_{k=1}^{\infty} B_k$ and

$$\sum_{k=1}^{\infty} \mathrm{vol}(B_k) \leq m^*(A) + \varepsilon.$$

We know that

$$m^*(B_k) = m^*(B_k \cap E) + m^*(B_k \setminus E) \text{ by Exercise 2.4.1.}$$

Thus,

$$\mathrm{vol}(B_k) = m^*(B_k \cap E) + m^*(B_k \setminus E).$$

This gives

$$\begin{aligned} \sum_{k=1}^{\infty} \mathrm{vol}(B_k) &= \sum_{k=1}^{\infty} m^*(B_k \cap E) + \sum_{k=1}^{\infty} m^*(B_k \setminus E) \geq m^*\left(\left(\bigcup_{k=1}^{\infty} B_k\right) \cap E\right) + m^*\left(\left(\bigcup_{k=1}^{\infty} B_k\right) \setminus E\right) \\ &\geq m^*(A \cap E) + m^*(A \setminus E). \end{aligned}$$

Hence,

$$m^*(A) + \varepsilon \geq \sum_{k=1}^{\infty} \mathrm{vol}(B_k) \geq m^*(A \cap E) + m^*(A \setminus E),$$

and since this is true for all $\varepsilon > 0$, so we're done. ■

Remark 2.4.2. A similar argument will also shows that any half-space of the form

$$\{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_j > 0\} \text{ or } \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_j < 0\}$$

for all $1 \leq j \leq n$ is measurable.

Lemma 2.4.2 (Properties of measurable sets).

(a) If E is measurable, then $\mathbb{R}^n \setminus E$ is also measurable.

(b) If E is measurable, and $x \in \mathbb{R}^n$, then $x + E$ is also measurable.

(c) If E_1 and E_2 are measurable, then $E_1 \cap E_2$ and $E_1 \cup E_2$ are measurable.

(d) If $E_1, \ldots, E_N$ are measurable, then $\bigcup_{j=1}^N E_j$ and $\bigcap_{j=1}^N E_j$ are measurable.

(e) Every open box and close box is measurable.

(f) Any set E of outer measure zero is measurable.

proof of (a). Since E is measurable, $m^*(A) = m^*(A \cap E) + m^*(A \setminus E)$ for any $A \subseteq \mathbb{R}^n$. Also,

$$A \cap (\mathbb{R}^n \setminus E) = A \setminus E \text{ and } A \setminus (\mathbb{R} \setminus E) = A \cap E.$$

Thus,

$$m^*(A) = m^*(A \setminus (\mathbb{R}^n \setminus E)) + m^*(A \cap (\mathbb{R}^n \setminus E))$$

for all $A \subseteq \mathbb{R}^n$, which means $\mathbb{R}^n \setminus E$ is measurable. ■

proof of (b). Suppose E is meausrable, then $m^*(B) = m^*(B \cap E) + m^*(B \setminus E)$ for any $B \subseteq \mathbb{R}^n$. To show $x + E$ is measurable, if we let $A \subseteq \mathbb{R}^n$ arbitrarily, we have

$$m^*(A - x) = m^*((A - x) \cap E) + m^*((A - x) \setminus E).$$

Now since $m^*(A) = m^*(A - x)$, and

$$(A - x) \cap E = (A \cap (E + x)) - x \text{ and } (A - x) \setminus E = (A \setminus (E + x)) - x,$$

we have

$$\begin{aligned} m^*(A) = m^*(A - x) &= m^*((A - x) \cap E) + m^*((A - x) \setminus E) \\ &= m^*((A \cap (E + x)) - x) + m^*((A \setminus (E + x)) - x) = m^*(A \cap (E + x)) + m^*(A \setminus (E + x)). \end{aligned}$$

Thus, $E + x$ is measurable. ■

proof of (c). It sufficies to show that $E_1 \cup E_2$ is measurable. Then, by Demorgan's law,

$$E_1 \cap E_2 = \mathbb{R}^n \setminus ((\mathbb{R}^n \setminus E_1) \cup (\mathbb{R}^n \setminus E_2)),$$

so we can show $E_1 \cap E_2$ is measurable. Now we want to show

$$m^*(A) = m^*(A \cap (E_1 \cup E_2)) + m^*(A \setminus (E_1 \cup E_2)).$$

Since E_1 is measurable,

$$m^*(A) = m^*(A \cap E_1) + m^*(A \setminus E_1).$$

Next, since E_2 is measurable,

$$m^*(A \setminus E_1) = m^*((A \setminus E_1) \cap E_2) + m^*((A \setminus E_1) \setminus E_2).$$

Hence,

$$m^*(A) = m^*(A \cap E_1) + m^*(A \setminus E_1) = m^*(A \cap E_1) + m^*((A \setminus E_1) \cap E_2) + m^*((A \setminus E_1) \setminus E_2).$$

Notice that $(A \setminus E_1) \cap E_2 = A \cap (E_2 \setminus E_1)$ and $(A \setminus E_1) \setminus E_2 = A \setminus (E_1 \cup E_2)$. Thus,

$$m^*(A) = m^*(A \cap E_1) + m^*(A \cap (E_2 \setminus E_1)) + m^*(A \setminus (E_1 \cup E_2)).$$

Note that

$$(A \cap E_1) \cup (A \cap (E_2 \setminus E_1)) = A \cap (E_1 \cup E_2),$$

so

$$m^*(A \cap (E_1 \cup E_2)) \leq m^*(A \cap E_1) + m^*(A \cap (E_2 \setminus E_1)).$$

This gives

$$m^*(A) \geq m^*(A \cap (E_1 \cup E_2)) + m^*(A \setminus (E_1 \cup E_2)).$$

On the other hand, we have $A = (A \cap (E_1 \cup E_2)) \cup (A \setminus (E_1 \cup E_2))$, so we have

$$m^*(A) = m^*(A \cap (E_1 \cup E_2)) + m^*(A \setminus (E_1 \cup E_2)).$$

■

proof of (d). This can be proved by induction on N with (c). ■

proof of (e). Note that every open box is the intersection of many half-spaces of the form

$$\{x : x_j > a_j\} \text{ or } \{x : x_j < b_j\}.$$

Thus, by Lemma 2.4.1 and (d), we know every open box is measurable. Closed boxes are complement of union of open half spaces and thus measurable. ■

proof of (f). Suppose $m^*(E) = 0$. For any $A \subseteq \mathbb{R}^n$, we have $A \cap E \subseteq E$, so

$$m^*(A \cap E) \leq m^*(E) = 0.$$

Hence,

$$m^*(A \cap E) + m^*(A \setminus E) = m^*(A \setminus E) \leq m^*(A).$$

Since

$$m^*(A \cap E) + m^*(A \setminus E) \geq m^*(A)$$

is trivial, so we're done. ■

Lemma 2.4.3 (Finite additivity). If $(E_j)_{j \in J}$ are a finite collection of disjoint measurable sets, then for any set A (not necessarily measurable), we have

$$m^*\left(A \cap \left(\bigcup_{j \in J} E_j\right)\right) = \sum_{j \in J} m^*(A \cap E_j).$$

Furthermore, we have

$$m\left(\bigcup_{j \in J} E_j\right) = \sum_{j \in J} m(E_j).$$

Proof. It sufficies to show the case of two sets: If E and F are disjoint measurable sets, then for any $A \subseteq \mathbb{R}^n$,

$$m^*(A \cap (E \cup F)) = m^*(A \cap E) + m^*(A \cap F).$$

Assume the statement is true for k pairwise disjoint measurable sets, and let $E_1, E_2, \ldots, E_k, E_{k+1}$ be pairwise disjoint measurable sets. Applying the two sets case,

$$m^*(A \cap (\bigcup_{i=1}^{k} E_i \cup E_{k+1})) = m^*(A \cap \bigcup_{i=1}^{k} E_i) + m^*(A \cap E_{k+1}).$$

Now the induction hypothesis gives

$$m^*(A \cap \bigcup_{i=1}^{k} E_i) = \sum_{i=1}^{k} m^*(A \cap E_i),$$

so

$$m^*(A \cap \left(\bigcup_{i=1}^{k+1} E_i\right)) = \sum_{i=1}^{k+1} m^*(A \cap E_i).$$

Now we show the two set cases. Suppose E and F are measurable and disjoint, then $U := E \cup F$ is measurable, and since E is measurable,

$$m^*(A \cap U) = m^*((A \cap U) \cap E) + m^*((A \cap U) \setminus E).$$

Note that

$$A \cap U \cap E = A \cap (E \cup F) \cap E = A \cap E \text{ and } (A \cap U) \setminus E = (A \cap (E \cup F)) \setminus E = A \cap F$$

since $E \cap F = \varnothing$. Hence,

$$m^*(A \cap U) = m^*(A \cap E) + m^*(A \cap F),$$

and we're done.

Now if we have pairwise disjoint $E_1, E_2, \ldots, E_k$, we can pick $A = \mathbb{R}^n$, and thus

$$m(\bigcup_{j \in J} E_j) = \sum_{j \in J} m(E_j).$$

■

Corollary 2.4.1. If $A \subseteq B$ and A, B ar both measurable, then $B \setminus A$ is measurable and

$$m(B \setminus A) + m(A) = m(B).$$

Proof. Since

$$B \setminus A = B \cap (\mathbb{R}^n \setminus A),$$

and A, B are measurable, so we can deduce $B \setminus A$ is measurable. Also, since $A \cap (B \setminus A) = \varnothing$, so by Lemma 2.4.3, we know

$$m(B) = m(A) + m(B \setminus A).$$

■

Lemma 2.4.4 (Countable additivity)**.** Let $(E_j)_{j \in J}$ be a countable collection of pairwise disjoint measurable subsets of $\mathbb{R}^n$, then

$$E := \bigcup_{j \in J} E_j$$

is measurable, and

$$m(E) = \sum_{j \in J} m(E_j).$$

Proof. Since J is countable, we may enumerate it as $\{j_1, j_2, \ldots\}$. For each $N \in \mathbb{N}$, define $F_N = \bigcup_{k=1}^N E_{j_k}$. Thus, F_N is measurable. By previous result,

$$m(F_N) = \sum_{k=1}^N m(E_{j_k}).$$

Let $E = \bigcup_{k=1}^\infty E_{j_k}$, we have $E = \bigcup_{N=1}^\infty F_N$ and $F_1 \subseteq F_2 \subseteq \ldots$.

We first show E is measurable. It sufficies to show that

$$m^*(A \cap E) + m^*(A \setminus E) \leq m^*(A) \text{ for all } A \subseteq \mathbb{R}^n.$$

Since F_N is measurable and $F_N \subseteq E$, we have

$$m^*(A) = m^*(A \cap F_N) + m^*(A \setminus F_N)$$

and

$$A \setminus E \subseteq A \setminus F_N \Rightarrow m^*(A \setminus F_N) \geq m^*(A \setminus E).$$

Also,

$$m^*(A \cap F_N) = \sum_{k=1}^N m^*(A \cap E_{j_k}).$$

Hence,

$$m^*(A) \geq \sum_{k=1}^{N} m^*(A \cap E_{j_k}) + m^*(A \setminus E) \text{ true for all } N.$$

Thus,

$$m^*(A) \geq \sum_{k=1}^{\infty} m^*(A \cap E_{j_k}) + m^*(A \setminus E).$$

Since

$$\bigcup_{k=1}^{\infty} (A \cap E_{j_k}) = A \cap \left(\bigcup_{k=1}^{\infty} E_{j_k} \right) = A \cap E,$$

we have

$$m^*(A \cap E) \leq \sum_{k=1}^{\infty} m^*(A \cap E_{j_k}),$$

so

$$m^*(A) \geq m^*(A \cap E) + m^*(A \setminus E).$$

This shows E is measurable.

Now we show

$$m(E) = \sum_{k=1}^{\infty} m(E_{j_k}).$$

First, since $E = \bigcup_{i=1}^{\infty} E_{j_k}$, so

$$m(E) \leq \sum_{k=1}^{\infty} m(E_{j_k}).$$

Now since $F_N \subseteq E$ for all N, so

$$\sum_{k=1}^{N} m(E_{j_k}) = m(F_N) \leq m(E),$$

so

$$m(E) = \sum_{k=1}^{\infty} m(E_{j_k}).$$

■

Lemma 2.4.5 (σ-algebra property)**.** If $(\Omega_j)_{j \in J}$ is a countable collection of measurable sets, then $\bigcup_{j \in J} \Omega_j$ and $\bigcap_{j \in J} \Omega_j$ are both measurable.

Proof. Write $J = \{j_1, j_2, \dots\}$ and define $E_1 = \Omega_{j_1}$ and

$$E_N = \Omega_N \setminus \left(\bigcup_{k=1}^{N-1} \Omega_{j_k} \right) \text{ for all } N \geq 2.$$

Hence, E_i is measurable for all $i \in \mathbb{N}$ and disjoint by construction. Note that

$$\bigcup_{k=1}^{\infty} E_k = \bigcup_{k=1}^{\infty} \Omega_{j_k}.$$

By this, $\bigcup_{k=1}^{\infty} \Omega_{j_k}$ is measurable since we have shown that the union of countably infinitely many pairwise disjoint measurable sets is measurable.

Now since

$$\bigcap_{k=1}^{\infty} \Omega_{j_k} = \mathbb{R}^n \setminus \bigcup_{k=1}^{\infty} (\mathbb{R}^n \setminus \Omega_{j_k}),$$

we know $\bigcap_{k=1}^{\infty} \Omega_{j_k}$ is measurable.

■

Lemma 2.4.6. Every open set can be written as a countable or finite union of open boxes.

Proof. We say an open box is rational if $B = \prod_{i=1}^{n}(a_i, b_i)$ where $a_i, b_i \in \mathbb{Q}$. The set of rational open boxes is countable (since this set is isomorphic to a subset of $\mathbb{Q}^{2n}$).

We make the following claim: given any open ball $B(x, r)$, $\exists$ a rational box B with $x \in B \subseteq B(x, r)$. To show this, let $x = (x_1, \ldots, x_n)$ by any point in $\mathbb{R}^n$. Let a_i, b_i be rational numbers s.t.

$$x_i - \frac{r}{n} < a_i < x_i < b_i < x_i + \frac{r}{n},$$

then we show that $B \subseteq B(x, r)$. Let $y \in B$, then

$$x_i - \frac{r}{n} < a_i < y_i < b_i < x_i + \frac{r}{n},$$

so

$$|y_i - x_i| < \frac{r}{n}.$$

Hence,

$$\|y - x\| = \sqrt{\sum_{i=1}^{n}(y_i - x_i)^2} \leq \sum_{i=1}^{n}|y_i - x_i| < \frac{r}{n} \cdot n = r.$$

Hence, $y \in B(x, r)$. Thus, we're done.

Now let E be an open set in $\mathbb{R}^n$. For any $x \in E$, $\exists r_x > 0$ s.t. $B(x, r_x) \subseteq E$. Thus, $E = \bigcup_{x \in E} B(x, r_x)$. Also, for any $B(x, r_x)$, there exists a rational box B_x s.t. $x \in B_x \subseteq B(x, r_x)$. Thus,

$$\bigcup_{x \in E} B_x = E.$$

■

Lemma 2.4.7. Every open set, and every closed set, is Lebesgue measurable.

Proof. By previous lemma, every open set can be written as the union of countable or finite rational open boxes, and every rational open box is measurable, so every open set is measurable. Since every closed box is $\mathbb{R}^n \setminus E$ for some open E, so it is also measurable. ■

Lecture 12

2.5 Measurable Functions

7 April.

Definition 2.5.1 (Measurable functions)**.** Let Ω be a measurable subset of $\mathbb{R}^n$, and let $f : \Omega \to \mathbb{R}^m$ be a function. Then, f is measurable iff $f^{-1}(V)$ is measurable for every open set $V \subseteq \mathbb{R}^m$, where

$$f^{-1}(V) = \{x \in \Omega \mid f(x) \in V\}$$

is the pre-image(or inverse image) of V under f.

Remark 2.5.1. In measure theory, this type of function is called Borel measurable.

Definition 2.5.2. Given a set X. A collection of subsets $\sigma(X)$ of X is called a σ-algebra if

(1) $X \in \sigma(X)$.

(2) $E \in \sigma(X)$ implies $X \setminus E \in \sigma(X)$.

(3) $E_i \in \sigma(X)$ for all $i \in \mathbb{N}$ implies $\bigcup_{i=1}^{\infty} E_i \in \sigma(X)$.

Remark 2.5.2. Since

$$\bigcap_{i=1}^{\infty} E_i = X \setminus \left(\bigcup_{i=1}^{\infty} (X \setminus E_i) \right),$$

so $E_i \in \sigma(X)$ for all $i \in \mathbb{N}$ implies $\bigcap_{i=1}^{\infty} E_i \in \sigma(X)$.

Definition 2.5.3. In $\mathbb{R}^n$, a set is called Borel measurable if it is in the smallest σ-algebra that contains all open sets.

Lemma 2.5.1. Let Ω be a measurable subset of $\mathbb{R}^n$, and $f : \Omega \to \mathbb{R}^m$ be continuous, then f is also measurable.

Proof. Let V be any open subset of $\mathbb{R}^m$, then since f is continuous, so $f^{-1}(V)$ is open, and since all open sets are measurable, we know $f^{-1}(V)$ is measurable, and thus f is measurable. ■

Lemma 2.5.2. Let Ω be a measurable subset of $\mathbb{R}^n$, and let $f : \Omega \to \mathbb{R}^m$ be a function. Then f is measurable if and only if $f^{-1}(B)$ is measurable for every open box B.

Proof.

($\Rightarrow$) If f is measurable, then since every open box B is open, $f^{-1}(B)$ is measurable.

($\Leftarrow$) Given any open set V, $\exists$ a countable union of open boxes s.t. $V = \bigcup_{\ell=1}^{\infty} B_\ell$ by Lemma 2.4.6, and

$$f^{-1}(V) = f^{-1}\left(\bigcup_{\ell=1}^{\infty} B_\ell \right) = \bigcup_{\ell=1}^{\infty} f^{-1}(B_\ell),$$

and since $f^{-1}(B_\ell)$ is measurable for all $\ell \in \mathbb{N}$, and thus $\bigcup_{\ell=1}^{\infty} f^{-1}(B_\ell)$ is measurable by Lemma 2.4.5.

■

Corollary 2.5.1. Let Ω be a measurable subset of $\mathbb{R}^n$, and let $f : \Omega \to \mathbb{R}^m$ be a function. Suppose $f = (f_1, f_2, \ldots, f_m)$, where $f_j : \Omega \to \mathbb{R}$ is the j-th coordinate of f. Then f is measurable if and only if f_j is measurable for all $1 \leq j \leq m$.

Proof.

($\Rightarrow$) Fix $j \in \{1, 2, \ldots, m\}$. Let $I \subseteq \mathbb{R}$ be an open interval. Consider the open set $B_j = \mathbb{R} \times \cdots \times I \times \cdots \times \mathbb{R}$, where I is in the j-th coordinate, then for $x \in \Omega$, we know

$$x \in f^{-1}(B_j) \Leftrightarrow f(x) \in B_j \Leftrightarrow f_j(x) \in I \Leftrightarrow x \in f_j^{-1}(I).$$

Hence, $f^{-1}(B_j) = f_j^{-1}(I)$. Since f is measurable, $f^{-1}(B_j)$ is measurable since B_j is an open box. Hence, $f_j^{-1}(I)$ is measurable, and since this is true for all open box I of $\mathbb{R}$, so by Lemma 2.5.2, f_j is measurable for all $1 \leq j \leq m$.

($\Leftarrow$) Let $B = I_1 \times I_2 \times \cdots \times I_m$ be an open box in $\mathbb{R}^m$. Then,

$$\begin{aligned} x \in f^{-1}(B) &\Leftrightarrow f(x) = (f_1(x), f_2(x), \ldots, f_m(x)) \in B = I_1 \times I_2 \times \cdots \times I_m \\ &\Leftrightarrow f_j(x) \in I_j \text{ for all } 1 \leq j \leq m \Leftrightarrow x \in f_j^{-1}(I_j) \text{ for all } 1 \leq j \leq m \\ &\Leftrightarrow x \in \bigcap_{j=1}^{m} f_j^{-1}(I_j). \end{aligned}$$

Hence, $f^{-1}(B) = \bigcap_{j=1}^{m} f_j^{-1}(I_j)$. Since for all $1 \leq j \leq m$, f_j is measurable and I_j is an open box in $\mathbb{R}$, so $f^{-1}(I_j)$ is measurable and thus $\bigcap_{j=1}^{m} f_j^{-1}(I_j)$ is measurable, so $f^{-1}(B)$ is measurable, and thus f is measurable.

■

Lemma 2.5.3. Let Ω be a measurable subset of $\mathbb{R}^n$, and let W be an open subset of $\mathbb{R}^m$. If $f : \Omega \to W$ is measurable and $g : W \to \mathbb{R}^p$ is continuous, then $g \circ f : \Omega \to \mathbb{R}^p$ is measurable.

Proof. Let V be an open set in $\mathbb{R}^p$, then $(g \circ f)^{-1}(V) = f^{-1}\left(g^{-1}(V)\right)$. Since g is continuous and V is open, $g^{-1}(V)$ is open in W, so $g^{-1}(V) = U \cap W$, where U is open in $\mathbb{R}^m$. Since W is also open in $\mathbb{R}^m$, $W \cap U$ is also open in $\mathbb{R}^m$, so $g^{-1}(V)$ is open in $\mathbb{R}^m$. Now since f is measurable, $f^{-1}\left(g^{-1}(V)\right)$ is measurable, i.e. $(g \circ f)^{-1}(V)$ is measurable. ■

Corollary 2.5.2. Let Ω be a measurable subset of $\mathbb{R}^n$. If $f : \Omega \to \mathbb{R}$ is measurable, then so is $|f|, \max\{f, 0\}$, and $\min\{f, 0\}$.

Proof. Since $g_1(x) := |x|, g_2(x) := \max\{x, 0\}$, and $g_3(x) := \min\{x, 0\}$ are all continuous, so by Lemma 2.5.3, $g_1 \circ f, g_2 \circ f, g_3 \circ f$ are all measurable.

Remark 2.5.3.

$$\max\{x, 0\} = \frac{x + |x|}{2}, \quad \min\{x, 0\} = \frac{x - |x|}{2}.$$

■

Corollary 2.5.3. Let Ω be a measurable subset of $\mathbb{R}^n$. If $f : \Omega \to \mathbb{R}$ and $g : \Omega \to \mathbb{R}$ are measurable, then so are $f + g, f - g, f \cdot g, \max\{f, g\}$, and $\min\{f, g\}$. If $g(x) \neq 0$ for all $x \in \Omega$, then f/g is also measurable.

Proof. Consider the map $F : \Omega \to \mathbb{R}^2$ defined by $F(x) = (f(x), g(x))$, then by Corollary 2.5.1, F is also measurable. Define $h_+(u, v) = u + v$, where $h_+ : \mathbb{R}^2 \to \mathbb{R}$ is continuous. Then $h_+ \circ F$ is measurable, where $(h_+ \circ F)(x) = f(x) + g(x)$. Similarly, we can define

$$\begin{aligned} h_*(u, v) &= uv, \\ h_{\max}(u, v) &= \max(u, v) = \frac{u + v + |u - v|}{2}, \\ h_{\min}(u, v) &= \min(u, v) = \frac{u + v - |u - v|}{2}, \end{aligned}$$

and they are all continuous, so $h_* \circ F, h_{\max} \circ F, h_{\min} \circ F$ are all measurable. As for the division, it is similar. ■

Lemma 2.5.4. Let Ω be a measurable subset of $\mathbb{R}^n$, and let $f : \Omega \to \mathbb{R}$ be a function, then f is measurable iff $f^{-1}((a, \infty))$ is measurable for any $a \in \mathbb{R}$.

Proof.

($\Rightarrow$) Since (a, ∞) is open, so $f^{-1}((a, \infty))$ is measurable.

($\Leftarrow$) Let I be an open interval in $\mathbb{R}$ with $I = (\alpha, \beta)$ where $\alpha < \beta$, then it sufficies to show $f^{-1}(I)$ is measurable by Lemma 2.5.2. Note that $(\alpha, \beta) = (\alpha, \infty) \cap (\mathbb{R} \setminus [\beta, \infty))$, and

$$[\beta, \infty) = \bigcap_{m=1}^{\infty} \left(\beta - \frac{1}{m}, \infty\right).$$

Therefore,

$$f^{-1}([\beta, \infty)) = f^{-1}\left(\bigcap_{m=1}^{\infty} \left(\beta - \frac{1}{m}, \infty\right)\right) = \bigcap_{m=1}^{\infty} f^{-1}\left(\left(\beta - \frac{1}{m}, \infty\right)\right),$$

which is measurable since $f^{-1}(a, \infty)$ is measurable for all $a \in \mathbb{R}$. Hence,

$$f^{-1}(\alpha, \beta) = f^{-1}(\alpha, \infty) \cap \left(\Omega \setminus f^{-1}(\beta, \infty)\right)$$

is measurable.

■

Definition 2.5.4 (Measurable functions in the extended reals)**.** Let Ω be a measurable subset of $\mathbb{R}^n$. Set

$$\mathbb{R}^* := \mathbb{R} \cup \{+\infty\} \cup \{-\infty\}.$$

A function $f : \Omega \to \mathbb{R}^*$ is said to be measurable if $f^{-1}((a, \infty])$ is measurable for all $a \in \mathbb{R}$, where

$$f^{-1}((a, \infty]) = f^{-1}((a, \infty)) \cup f^{-1}(\{\infty\}).$$

Lecture 13

9 April.

Example 2.5.1. Suppose $f(x) = \frac{1}{x^2}$ is defined on $\mathbb{R}$, then we can extend it to $f : \mathbb{R} \to \mathbb{R}^*$ so that

$$f(x) = \begin{cases} \dfrac{1}{x^2}, & \text{if } x \neq 0; \\ \infty, & \text{if } x = 0. \end{cases}$$

Then f is measurable. Note that if we define f by $f : \Omega \to \mathbb{R}$, then $f^{-1}((a, \infty]) = f^{-1}((a, \infty))$ for all $a \in \mathbb{R}$.

Lemma 2.5.5. Let $\Omega \subseteq \mathbb{R}^n$ be measurable. For each positive integer n, let $f_n : \Omega \to \mathbb{R}^*$ be a measurable function. Then, $\sup_{n\in\mathbb{N}} f_n, \inf_{n\in\mathbb{N}} f_n, \limsup_{n\to\infty} f_n, \liminf_{n\to\infty} f_n$ are measurable. In particular, if $\lim_{n\to\infty} f_n(x) = f(x)$ for $f : \Omega \to \mathbb{R}^*$, then f is also measurable.

Proof. Let $g(x) = \sup_{n\in\mathbb{N}} f_n(x)$, then g is measurable iff $g^{-1}((a, \infty])$ is measurable for any $a \in \mathbb{R}$. Also,

$$x \in g^{-1}((a, \infty]) \Leftrightarrow g(x) > a \Leftrightarrow \sup_{n\in\mathbb{N}} f_n(x) > a \Leftrightarrow \exists k \in \mathbb{N} \text{ s.t. } f_k(x) > a$$
$$\Leftrightarrow \exists k \in \mathbb{N} \text{ s.t. } x \in f_k^{-1}((a, \infty]) \Leftrightarrow x \in \bigcup_{n=1}^{\infty} f_n^{-1}((a, \infty]).$$

Hence, $g^{-1}((a, \infty]) = \bigcup_{n=1}^{\infty} f_n^{-1}((a, \infty])$, so it sufficies to show that $\bigcup_{n=1}^{\infty} f_n^{-1}((a, \infty])$ is measurable. However, since for all $n \in \mathbb{N}$, f_n is measurable, so $f_n^{-1}((a, \infty])$ is measurable, and thus we're done.

Let $h(x) = \inf_{n\in\mathbb{N}} f_n(x)$, then h is measurable iff $h^{-1}((a, \infty])$ is measurable for all $a \in \mathbb{R}$. Now

since

$$x \in h^{-1}((a,\infty]) \Leftrightarrow h(x) \in (a,\infty] \Leftrightarrow \inf_{n\in\mathbb{N}} f_n(x) \in (a,\infty] \Leftrightarrow \inf_{n\in\mathbb{N}} f_n(x) > a \Leftrightarrow \forall n \in \mathbb{N},\ f_n(x) > a$$

$$\Leftrightarrow \forall n \in \mathbb{N},\ f_n(x) \in (a,\infty] \Leftrightarrow \forall n \in \mathbb{N},\ x \in f_n^{-1}((a,\infty]) \Leftrightarrow x \in \bigcap_{n=1}^{\infty} f_n^{-1}((a,\infty]).$$

Hence, $h^{-1}((a,\infty]) = \bigcap_{n=1}^{\infty} f_n^{-1}((a,\infty])$. However, since for all $n \in \mathbb{N}$, f_n is measurable, so $f_n^{-1}((a,\infty])$ is measurable, and thus we're done.

Now we define $g_N(x) = \sup_{n\geq N} f_n(x)$, then we know g_N is measurable for all $N \in \mathbb{N}$ by what we've proved. Now since

$$\limsup_{n\to\infty} f_n(x) = \lim_{n\to\infty} \sup_{k\geq n} f_k(x) = \inf_{N\geq 1} \sup_{n\geq N} f_n(x) = \inf_{N\geq 1} g_N(x),$$

so by what we've proved, we know $\inf_{N\geq 1} g_N(x)$ is measurable, and we're done.

Similar argument can be used to prove $\liminf_{n\to\infty} f_n(x)$ is measurable.

■

Chapter 3

Lebesgue Integration

3.1 What is Lebesgue integration?

Lebesgue integration is a way of defining the integral of a function by grouping together points that have *similar output values*. Instead of cutting the x-*axis* into many thin intervals, as in Riemann integration, Lebesgue integration studies the sets of points where the function lies in certain value ranges and measures the size of those sets.

Informally, the two viewpoints are:

- **Riemann integration:** slice the domain into vertical strips and add "width $\times$ height."
- **Lebesgue integration:** slice the range into horizontal bands and add "value $\times$ measure of the set having that value."

This change of perspective makes Lebesgue integration much more flexible. It can integrate many functions that are badly behaved from the Riemann point of view, and it works especially well in probability, Fourier analysis, and modern analysis.

The main idea

Suppose a function f takes the value approximately y on a set of points whose length is approximately m. Then its contribution to the integral is approximately

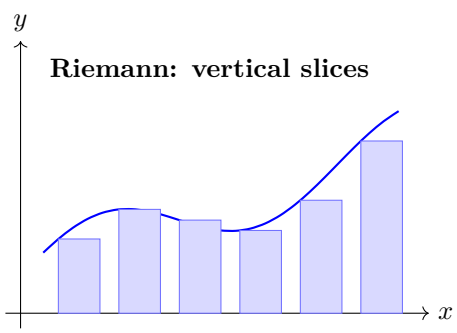
$$y \cdot m.$$

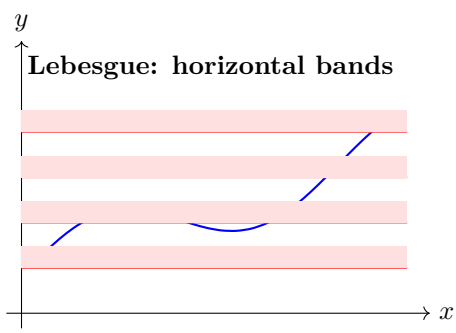
Adding these contributions over all output levels leads to the Lebesgue integral.

For nonnegative functions, one can think of the construction as approximating f by step functions built from *horizontal* layers. In this way, the Lebesgue integral measures area by organizing the graph according to function values rather than according to input intervals.

How is it different from Riemann integration?

The essential difference is the direction of the partition.





- In **Riemann integration**, the interval on the x-axis is divided into many small subintervals. One samples the function in each interval and sums rectangle areas.
- In **Lebesgue integration**, the y-axis is divided into small bands. For each band, one looks at all x such that $f(x)$ falls inside that band, then measures how large that set is.

So Riemann organizes the area by *where the input lies*, while Lebesgue organizes it by *what values the function takes*.

Why Lebesgue integration is more powerful

Lebesgue integration handles limits of functions much better. For example, if a sequence of functions converges in a reasonable way, the Lebesgue integral often allows us to interchange limit and integral. This is one of the main reasons it became the standard notion of integration in advanced mathematics.

It also integrates some functions that are not Riemann integrable. The classical example is the Dirichlet function on $[0, 1]$:

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

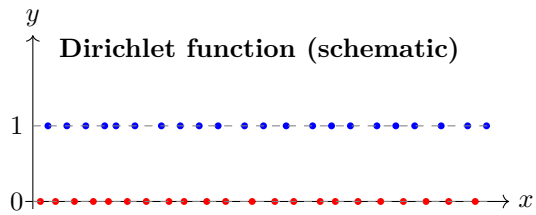
This function is discontinuous at every point, so it is not Riemann integrable. However, the rational numbers in $[0, 1]$ have Lebesgue measure 0, so from the Lebesgue point of view the function is equal to 0 almost everywhere. Therefore,

$$\int_0^1 f(x)\, dx = 0$$

in the Lebesgue sense.

A visual example

The next picture shows the Dirichlet function schematically. The dense blue points near $y = 1$ represent rational inputs, and the dense red points near $y = 0$ represent irrational inputs. Since the rational set has measure zero, the upper layer contributes no area in the Lebesgue sense.



Summary

- **Riemann integration** partitions the domain and adds areas of vertical rectangles.
- **Lebesgue integration** partitions the range and measures the size of the sets where the function takes certain values.
- Lebesgue integration is better suited to complicated functions and to taking limits.
- Every Riemann integrable function is Lebesgue integrable, but not every Lebesgue integrable function is Riemann integrable.

In short, Lebesgue integration is a more robust and more powerful notion of area, built to handle modern analysis.

Definition 3.1.1. Let $\Omega \subseteq \mathbb{R}^n$ be a measurable set, and let $f : \Omega \to \mathbb{R}$ be a measurable function. We say f is a simple function if the image of $f(\Omega)$ is finite, i.e. $\exists c_1, \ldots, c_N$ s.t. for any $c \in \Omega$, $f(x) = c_j$ for some $1 \leq j \leq N$.

Example 3.1.1. Let $\Omega \subseteq \mathbb{R}^n$ be a measurable set, and let E be a measurable subset of Ω. Define the characteristic function $\chi_E : \Omega \to \mathbb{R}$ by

$$\chi_E(x) = \begin{cases} 1, & \text{if } x \in E; \\ 0, & \text{if } x \in \Omega \setminus E. \end{cases}$$

Then, we claim that χ_E is measurable. Note that for any open interval I, we have

$$\chi_E^{-1}(I) = \begin{cases} \varnothing, & \text{if } 0 \notin I, 1 \notin I; \\ E, & \text{if } 0 \notin I, 1 \in I; \\ \Omega \setminus E, & \text{if } 0 \in I, 1 \notin I; \\ \Omega, & \text{if } 0 \in I, 1 \in I. \end{cases}$$

Thus, χ_E is measurable.

Lemma 3.1.1. Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and let $f : \Omega \to \mathbb{R}$ and $g : \Omega \to \mathbb{R}$ be simple function. Then, $f + g$ is also simple and for any $c \in \mathbb{R}$, the function cf is also simple.

Proof. Suppose $f(\Omega) = \{a_1, \ldots, a_N\}$ and $g(\Omega) = \{b_1, \ldots, b_M\}$, then $(f+g)(\Omega) = \{a_i + b_j\}_{\substack{1 \leq i \leq N \\ 1 \leq j \leq M}}$, which is finite, so $f + g$ is simple.

Also, $(cf)(\Omega) = \{ca_1, \ldots, ca_N\}$, which is also finite, so cf is simple. ■

Lemma 3.1.2. Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and let $f : \Omega \to \mathbb{R}$ be a simple function. Then, there exists finitely many distinct real numbers $c_1, \ldots, c_N$ and finitely many pairwise disjoint measurable sets $E_1, \ldots, E_N \subseteq \Omega$ s.t. $f = \sum_{j=1}^N c_j \chi_{E_j}$.

Proof. Suppose $f(\Omega) = \{c_1, \ldots, c_N\}$, where $c_i \neq c_j$ for all $i \neq j$. Define $E_j = f^{-1}(\{c_j\})$ for each $1 \leq j \leq N$, then $(E_j)_{j=1}^N$ are pairwise disjoint. Since $f(x) = c_\ell$ for some $1 \leq \ell \leq N$ for all $x \in \Omega$, $\Omega = \bigcup_{j=1}^N E_j$. Note that

$$E_j = f^{-1}(\{c_j\}) = \bigcap_{k=1}^{\infty} f^{-1}\left(\left(c_j - \frac{1}{k}, c_j + \frac{1}{k}\right)\right),$$

then since f is measurable, we know E_j is measurable. Now for $x \in \Omega$, we know $x \in E_k$ for some $1 \leq k \leq N$. Hence, $f(x) = c_k$, and thus

$$\sum_{j=1}^N c_j \chi_{E_j}(x) = c_k \chi_{E_k}(x) = c_k = f(x).$$

Thus, $f = \sum_{j=1}^N c_j \chi_{E_j}$. ■

Lemma 3.1.3. Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and let $f : \Omega \to [0, \infty]$ be a measurable function. Then there exists a sequence of simple functions $\{f_n\}_{n=1}^\infty$ with $f_n : \Omega \to \mathbb{R}$ s.t. each $f_n \geq 0$, and the sequence is increasing:

$$0 \leq f_1(x) \leq f_2(x) \leq \ldots \qquad \text{for all } x \in \Omega,$$

and $\lim_{n \to \infty} f_n(x) = f(x)$ for all $x \in \Omega$.

Proof. For each integer $n \geq 1$, define

$$E_{n,k} := \left\{ x \in \Omega : \frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n} \right\}, \quad k = 0, 1, \ldots, n2^n - 1,$$

and $F_n := \{x \in \Omega : f(x) \geq n\}$. Then define

$$f_n(x) := \sum_{k=0}^{n2^n-1} \frac{k}{2^n} \chi_{E_{n,k}}(x) + n\chi_{F_n}(x).$$

Thus, f_n is obtained from f by truncating at height n and then rounding down to the nearest multiple of 2^{-n}. Note that If $f(x) < n$, then $\exists 0 \leq k \leq n2^n - 1$ s.t.

$$\frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n},$$

so $f_n(x) = \frac{k}{2^n}$, and if $f(x) \geq n$, then $f_n(x) = n$.

Now we try to write down the explicit formula for f_n. Fix $x \in \Omega$, then write $a := f(x) \in [0, \infty]$.

Claim 3.1.1.

$$f_n(x) = \min\left(n, \frac{\lfloor 2^n a \rfloor}{2^n}\right) = \min\left(n, \frac{\lfloor 2^n f(x) \rfloor}{2^n}\right).$$

Proof. If $a = f(x) \geq n$, then $f_n(x) = n$, and $n \leq \frac{\lfloor 2^n f(x) \rfloor}{2^n}$, so this is true. If $a = f(x) < n$, then $\exists 0 \leq k \leq n2^n - 1$ s.t.

$$\frac{k}{2^n} \leq a < \frac{k+1}{2^n} \Leftrightarrow k \leq 2^n a < k + 1,$$

so $k = \lfloor 2^n a \rfloor$. Since $f(x) \in E_{n,k}$, we have

$$f_n(x) = \frac{k}{2^n} = \frac{\lfloor 2^n a \rfloor}{2^n}.$$

Thus, in this case, $f_n(x) = \min\left(n, \frac{\lfloor 2^n a \rfloor}{2^n}\right)$.

⊛

By this claim, we know $=0 \leq f_n(x) \leq f(x)$ for all n.

Now we show $f_n(x) \leq f_{n+1}(x)$.

- Case 1: $n \leq f(x)$, then $f_n(x) = n$, and $f_{n+1}(x) = \min\left(n+1, \frac{\lfloor 2^{n+1} f(x) \rfloor}{2^{n+1}}\right) \geq n$, so we're done.
- Case 2: $f(x) < n$, then

$$f_n(x) = \min\left(n, \frac{\lfloor 2^n f(x) \rfloor}{2^n}\right) = \frac{\lfloor 2^n f(x) \rfloor}{2^n}.$$

Thus, $\exists$ unique $m = \lfloor 2^n f(x) \rfloor \leq 2^n f(x) < m + 1$, and

$$f_{n+1}(x) = \min\left(n+1, \frac{\lfloor 2^{n+1} f(x) \rfloor}{2^{n+1}}\right),$$

and note that

$$2m \leq 2^{n+1} f(x) < 2(m+1),$$

so

$$f_n(x) = \frac{\lfloor 2^n f(x) \rfloor}{2^n} = \frac{m}{2^n} = \frac{2m}{2^{n+1}} \leq \frac{\lfloor 2^{n+1} f(x) \rfloor}{2^{n+1}} = f_{n+1}(x),$$

where the last equality holds since $f(x) < n < n + 1$.

Now we prove $\lim_{n\to\infty} f_n(x) = f(x)$.

- Case 1: $f(x) = \infty$, then $f_n(x) = n$, so $\lim_{n\to\infty} f_n(x) = x$.
- Case 2: $f(x) < \infty$, then $\exists N \in \mathbb{N}$ s.t. $f(x) < N$. If $n \geq N$, then

$$f_n(x) = \frac{k}{2^n} \text{ where } \frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n}.$$

Thus, $f_n(x) \leq f(x) \leq f_n(x) + \frac{1}{2^n}$, i.e.

$$|f(x) - f_n(x)| < \frac{1}{2^n}.$$

Thus, $\lim_{n\to\infty} f_n(x) = f(x)$.

■

Lecture 14

21 April.

Definition 3.1.2. Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and let $f : \Omega \to \mathbb{R}$ be a simple function which is non-negative, thus f is measurable and the image $f(\Omega)$ is finite and contained in $[0, \infty)$. Thus, we can define the Lebesgue integral $\int_\Omega f$ of f on Ω by

$$\int_\Omega f = \sum_{\lambda \in f(\Omega), \lambda > 0} \lambda m\left(\{x \in \Omega : f(x) = \lambda\}\right) = \sum_{\lambda \in f(\Omega)} \lambda m\left(\{x \in \Omega : f(x) = \lambda\}\right).$$

We'll also sometimes write $\int_\Omega f$ as $\int_\Omega f \, dm$.

Remark 3.1.1. By definition, we know simple function can be written as

$$f = \sum_{j=1}^{N} c_j \chi_{E_j},$$

where $\{c_1, \ldots, c_N\}$ are distinct and $\{E_j\}_{j=1}^N$ are disjoint, measurable subsets of Ω defined by $E_j = f^{-1}(\{c_j\})$, and thus

$$\int_\Omega f = \sum_{j=1}^{N} c_j m(E_j).$$

Example 3.1.2. Let $f : \mathbb{R} \to \mathbb{R}$ be the function which is equal to 3 on $[1, 2]$ and equal to 4 on $(2, 4)$ and is zero elsewhere. Then,

$$\int_\mathbb{R} f = 3m([1, 2]) + 4m([2, 4]) = 11.$$

Example 3.1.3. Suppose $g : \mathbb{R} \to \mathbb{R}$ is the function which is 1 on $[0, \infty)$ and is zero on $(-\infty, 0)$, then $g = 1 \cdot \chi_{[0.\infty)}$ and

$$\int_\mathbb{R} g = 1 \cdot m([0, \infty)) = \infty.$$

Lemma 3.1.4. Let $\Omega \subseteq \mathbb{R}^n$ be measurable and let $E_1, E_2, \ldots, E_N$ be pairwise disjoint measurable

subset of Ω. Let $c_1, \ldots, c_N$ be non-negative real numbers (not required to be distinct). Then,

$$\int_\Omega \sum_{j=1}^N c_j \chi_{E_j} = \sum_{j=1}^N c_j m(E_j).$$

Proof. Define $f := \sum_{j=1}^N c_j \chi_{E_j}$, then f is simple by definition. Let $d_1, \ldots, d_L$ be the distinct positive numbers that occurs among $c_1, \ldots, c_N$. Let

$$F_\ell := \bigcup_{\{j : c_j = d_\ell\}} E_j.$$

Then since each E_j is measurable, so F_ℓ is measurable. Note that $F_1, F_2, \ldots, F_\ell$ are pairwise disjoint since $E_1, \ldots, E_N$ are pairwise disjoint. Also, note that

$$f = \sum_{\ell=1}^L d_\ell \chi_{F_\ell}.$$

Now since $d_1, \ldots, d_L$ are distinct and $F_1, F_2, \ldots, F_L$ are pairwise disjoint, by definition we have

$$\int_\Omega f = \sum_{\ell=1}^L d_\ell m(F_\ell) = \sum_{\ell=1}^L d_\ell \sum_{j : c_j = d_\ell} m(E_j) = \sum_{\ell=1}^L \sum_{j : c_j = d_\ell} c_j m(E_j) = \sum_{j=1}^N c_j m(E_j).$$

■

Proposition 3.1.1. Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and let $f : \Omega \to \mathbb{R}$ and $g : \Omega \to \mathbb{R}$ be non-negative simple functions.

(a) We have $0 \leq \int_\Omega \leq \infty$. Furthermore, we have $\int_\Omega f = 0$ iff $m(\{x \in \Omega : f(x) \neq 0\}) = 0$.

(b) $\int_\Omega f + g = \int_\Omega f + \int_\Omega g$.

(c) For any $c > 0$, we have $\int_\Omega cf = c \int_\Omega f$.

(d) If $f(x) \leq g(x)$ for all $x \in \Omega$, then $\int_\Omega f \leq \int_\Omega g$.

Remark 3.1.2. We make a very convenient notational convention: if a property $P(x)$ holds for all points in Ω, except for a set of measure zero, then we say that P holds for almost every point in Ω. Thus (a) asserts that $\int_\omega f = 0$ if and only if f is zero for almost every point in Ω.

Proof. Since

$$f = \sum_{\lambda \in f(\Omega) \setminus \{0\}} \lambda \chi_{x \in \Omega : f(x) = \lambda},$$

we can write $f = \sum_{j=1}^N c_j \chi_{E_j}$, where $E_1, \ldots, E_N$ are pairwise disjoint, and each of c_j is positive. Similarly, we write $g = \sum_{j=1}^M d_j \chi_{F_j}$, where $F_1, \ldots, F_M$ are pairwise disjoint and $d_1, \ldots, d_M$ are positive.

(a) Since $\int_\Omega f = \sum_{j=1}^N c_j m(E_j)$, so it is clear that $0 \leq \int_\Omega f \leq \infty$. If f is zero almost everywhere, then all of E_j must have measure 0 (because $E_j \subseteq \{x \in \Omega : f(x) \neq 0\}$), and thus $\int_\Omega f = 0$.

Conversely, if $\int_\Omega f = 0$, then $\sum_{j=1}^N c_j m(E_j) = 0$, which can only happen when all of $m(E_j)$ are 0, and hence f is zero almost everywhere.

(b) Set

$$E_0 = \Omega \setminus \bigcup_{j=1}^N E_j, \quad c_0 = 0,$$

then $E_0, E_1, \ldots, E_N$ are pairwise disjoint, and their union is Ω, and $f = \sum_{j=0}^{N} c_j \chi_{E_j}$. Similarly, we can define

$$F_0 = \Omega \setminus \bigcup_{k=1}^{M} F_k, \quad d_0 = 0,$$

and $g = \sum_{k=0}^{M} d_k \chi_{F_k}$, where $F_0, F_1, \ldots, F_M$ are pairwise disjoint and their union is Ω. Now we know

$$\{E_j \cap F_k : 0 \leq j \leq N, 0 \leq k \leq M\}$$

forms a partition of Ω. Now since

$$\chi_{E_j} = \sum_{k=0}^{M} \chi_{E_j \cap F_k}, \quad \chi_{F_k} = \sum_{j=0}^{N} \chi_{E_j \cap F_k},$$

we have

$$f = \sum_{j=0}^{N} \sum_{k=0}^{M} c_j \chi_{E_j \cap F_k}, \quad g = \sum_{j=0}^{N} \sum_{k=0}^{M} d_k \chi_{E_j \cap F_k}.$$

Hence,

$$f + g = \sum_{j=0}^{N} \sum_{k=0}^{M} (c_j + d_k) \chi_{E_j \cap F_k},$$

which shows

$$\int_\Omega (f+g) = \sum_{j=0}^{N} \sum_{k=0}^{M} (c_j + d_k) m(E_j \cap F_k) = \int_\Omega f + \int_\Omega g$$

by previous lemma since

$$\int_\Omega f = \sum_{j=0}^{N} c_j m(E_j) = \sum_{j=0}^{N} \sum_{k=0}^{M} c_j m(E_j \cap F_k),$$

and similarly we have

$$\int_\Omega g = \sum_{k=0}^{M} d_k m(F_k) = \sum_{k=0}^{M} \sum_{j=0}^{N} d_k m(E_j \cap F_k).$$

(c) Since $cf = \sum_{j=1}^{N} (cc_j) \chi_{E_j}$, we have

$$\int_\Omega cf = \sum_{j=1}^{N} cc_j m(E_j) = c \int_\Omega f.$$

(d) Write $h \coloneqq g - f$, then h is simple and $g = h + f$. Hence, by (b), $\int_\Omega g = \int_\Omega h + \int_\Omega f$, but by (a) $\int_\Omega h \geq 0$, so we're done.

■

3.2 Integration of non-negative measurable functions

We now pass from integrating non-negative simple functions to integrating general non- negative measurable functions. In this section we allow our measurable functions to take the value $+\infty$.

Definition 3.2.1. Let $f : \Omega \to \mathbb{R}$ and $g : \Omega \to \mathbb{R}$ be functions, we say that f majorizes g, or

equivalently that g minorizes f, if

$$f(x) \geq g(x) \quad \forall x \in \Omega.$$

We sometimes use the phrase "f dominates g"instead of "f majorizes g". We also use the phrase g minorizes f.

Definition 3.2.2. Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and $f : \Omega \to [0, \infty]$ be measurable and non-negative. We define the Lebesgue integral $\int_\Omega f$ of f on Ω by

$$\int_\Omega f := \sup \left\{ \int_\Omega s : s \text{ is simple and non-negative, and } s \text{ minorizes } f \right\}.$$

Remark 3.2.1. If $\Omega' \subseteq \Omega$ is a measurable subset of Ω, then

$$\int_{\Omega'} f = \int_{\Omega'} f|_{\Omega'}.$$

Remark 3.2.2. $0 \leq \int_\Omega f \leq \infty$ since zero function is simple and non-negative, and minorizes f for all f.

Proposition 3.2.1. Let Ω be a measurable set, and let $f : \Omega \to [0, \infty]$ and $g : \Omega \to [0, \infty]$ be non-negative measurable functions.

(a) We have $0 \leq \int_\Omega f \leq \infty$. Furthermore, $\int_\Omega f = 0$ iff $f(x) = 0$ for almost every $x \in \Omega$.

(b) For any positive c, $\int_\Omega cf = c \int_\omega f$.

(c) If $f(x) \leq g(x)$ for all $x \in \Omega$, then $\int_\Omega f \leq \int_\Omega g$.

(d) If $f(x) = g(x)$ for all $x \in \Omega$, then $\int_\Omega f = \int_\Omega g$.

(e) If $\Omega' \subseteq \Omega$ is measurable, then

$$\int_{\Omega'} f = \int_\Omega f \chi_{\Omega'} \leq \int_\Omega f.$$

Appendix

Appendix A

Midterm Practice

A.1 HW6

Problem A.1.1. Let A be an open interval in $\mathbb{R}$. Show that

$$m^*(A) = m^*(A \cap (0, \infty)) + m^*(A \setminus (0, \infty)).$$

Proof. Let $A = (\alpha, \beta)$, then

$$A \cap (0, \infty) = \begin{cases} \varnothing, & \text{if } \beta \leq 0; \\ (\alpha, \beta), & \text{if } \alpha, \beta > 0; \\ (0, \beta), & \text{if } \alpha \leq 0, \beta > 0, \end{cases}$$

and

$$A \setminus (0, \infty) = \begin{cases} (\alpha, \beta), & \text{if } \beta \leq 0; \\ \varnothing, & \text{if } \alpha, \beta > 0; \\ (\alpha, 0], & \text{if } \alpha \leq 0, \beta > 0. \end{cases}$$

Thus, in each case

$$m^*(A) = m^*(A \cap (0, \infty)) + m^*(A \setminus (0, \infty)).$$

■

Problem A.1.2. Let A be an open box in $\mathbb{R}^n$, and let $E = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_n > 0\}$. Show that

$$m^*(A) = m^*(A \cap E) + m^*(A \setminus E).$$

Proof. Let $A = \prod_{i=1}^{\infty}(a_i, b_i)$, then

$$A \cap E = \left(\prod_{i=1}^{n-1}(a_i, b_i)\right) \times ((a_n \cap b_n) \cap (0, \infty)).$$

Now since $(a_n, b_n) \cap (0, \infty)$ is an interval,

$$m^*(A \cap E) = \operatorname{vol}(A_{n-1}) \cdot m^*(A_n \cap (0, \infty)) \text{ where } A_{n-1} = \prod_{i=1}^{n-1}(a_i, b_i),\ A_n = (a_n, b_n).$$

Similarly,

$$A \setminus E = A_{n-1} \times (A_n \setminus (0, \infty)),$$

so

$$m^*(A \setminus E) = \operatorname{vol}(A_{n-1}) \cdot m^*(A_n \setminus (0, \infty)).$$

Hence,

$$\begin{aligned} m^*(A) = \mathrm{vol}(A) = \mathrm{vol}(A_{n-1})(m^*(A_n)) &= \mathrm{vol}(A_{n-1})(m^*(A_n \cap (0,\infty)) + m^*(A_n \setminus (0,\infty))) \\ &= \mathrm{vol}(A_{n-1}) \cdot m^*(A_n \cap (0,\infty)) + \mathrm{vol}(A_{n-1}) \cdot m^*(A_n \setminus (0,\infty)) \\ &= m^*(A \cap (0,\infty)) + m^*(A \setminus (0,\infty)). \end{aligned}$$

■

A.2 HW1

Problem A.2.1. Let

$$D = \left\{(x,y) \in \mathbb{R}^n, x \geq 0, 0 \leq y \leq x^3\right\},$$

and $x_0 = (0,0)$. Define $f : D \to \mathbb{R}$ by $f(x,y) = 2x + \sqrt{y}$.

(a) Prove that f is differentiable at $(0,0)$ on D.

(b) Describe the set of all derivative $L(x,y)$ for f on $(0,0)$ on D.

A.3 HW2

Problem A.3.1. Define

$$\begin{aligned} \mathrm{GL}(n,\mathbb{R}) &= \{A \text{ is a } n \times n \text{ matrix and } A \text{ is invertible}\} \\ &= \{A \mid \det(A) \neq 0\}, \end{aligned}$$

which is open in $\mathbb{R}^{n+m}$. Consider the inverse map $g : \mathrm{GL}(n,r) \to \mathrm{GL}(n,r)$, i.e. $g(A) = A^{-1}$. Show that $g'(A) : \mathbb{R}^{n\times n} \to \mathrm{R}^{n\times n}$ is $g'(A)(H) = A^{-1}HA^{-1}$ for every $H \in \mathbb{R}^{n\times n}$.

Remark A.3.1. We can use the fact that if $M \in \mathbb{R}^{n+m}$ has $\|M\| < 1$, then

$$(I+M)^{-1} = I - M + M^2 - M^3 + \dots$$

converges in $\mathbb{R}^{n+m}$.

A.4 HW3

Problem A.4.1. Let $f : S \to S$ be a map between a complete metric space S. Assume there is a real sequence $\{\alpha_n\}$ which converges to 0 s.t.

$$d(f^n(x), f^n(y)) \leq \alpha_n d(x,y)$$

for all $n \geq 1$ and all $x, y \in S$. Prove that f has a unique fixed point. Hint. Apply the fixed-point theorem to f^m for a suitable m.

A.5 HW4

Problem A.5.1. Let $n \geq 2$, and let

$$f : \mathbf{R}^n \to \mathbf{R}$$

be a smooth function. Define

$$S := \{x = (x_1, \dots, x_n) \in \mathbf{R}^n : f(x) = 0\}.$$

Assume that

$$\nabla f(x) \neq 0 \qquad \text{for every } x \in S.$$

Then for every point $p \in S$, there exists an index $j \in \{1, \ldots, n\}$, an open neighborhood $W \subset \mathbf{R}^n$ of p, an open set $U \subset \mathbf{R}^{n-1}$, and a differentiable function

$$g : U \to \mathbf{R}$$

such that, after possibly reordering the coordinates, the set $S \cap W$ can be written in the form

$$S \cap W = \Big\{(x_1, \ldots, x_{j-1}, g(x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_n), x_{j+1}, \ldots, x_n) : (x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_n) \in U\Big\}.$$

In other words, near every point of S, the hypersurface S is locally the graph of a differentiable function of $n-1$ variables.

A.6 HW5

Problem A.6.1. Recall that a function $f : [a, b] \to \mathbf{R}$ is *Lipschitz* if there exists $L \geq 0$ such that $|f(x) - f(y)| \leq L|x - y|$ for all $x, y \in [a, b]$. Define the graph of f as:

$$\Gamma_f := \{(x, f(x)) : x \in [a, b]\} \subset \mathbf{R}^2.$$

Prove that $m_2^*(\Gamma_f) = 0$. *Hint: Partition $[a, b]$ into n sub-intervals and cover the corresponding pieces of the graph with rectangles. Consider the limit as $n \to \infty$.*

A.7 Random Stuff

Theorem A.7.1. If f is continuoius and I is an open interval, then $f(I)$ is also an interval.

Theorem A.7.2. Any interval in $\mathbb{R}$ is measurable (open, closed, or half-open, half-closed)

Theorem A.7.3. A set E is Lebesgue measurable iff $m^*(A) = m^*(A \cap E) + m^*(A \setminus E)$ for all $A \subseteq \mathbb{R}^n$. To show E is measurable, it suffices to show

$$m^*(A) \geq m^*(A \cap E) + m^*(A \setminus E).$$